Capture-Complete Learning for Transferable Bullet-Hell Control

Touhou Solver Research Project

Working draft — August 22, 2026

Abstract

We study control of the original Japanese Touhou 6 version 1.02h under Wine, with the scientific goal of completing a Lunatic no-miss, no-bomb route and eventually transferring the learning protocol to Touhou 8. The system is capture-complete but prediction-incomplete: it records coherent physical facts, filters only actions that intersect already-instantiated hazards over a fixed short horizon, and leaves action-relative future risk to a learned scorer. We separate the goal metric, no-miss completion probability, from the first optimization target, expected undiscounted physical HIT count over complete episodes. The latter is a dense surrogate and is not claimed to rank policies identically to the goal metric. The initial learner is deliberately limited to a causal decision-epoch dataset, a transparent supervised baseline, and immutable Wine evaluation. History encoders, offline value learning, uncertainty, adaptive collection, learned dynamics, and exact-prefix branching are admitted one at a time only after the preceding preregistered experiment exposes the concrete boundary that the added mechanism addresses. The paper reports an audited six-stage infrastructure baseline with 108 HITs, a completed causal learner-view audit, a negative serial/parallel equivalence gate, a completed negative Stage 4 supervised-baseline gate, and an inconclusive optimization-only follow-up. A second optimization-only follow-up reached its train-only convergence criterion but failed the unchanged calibration gate. The fitted current-observation behavior clones improved held-out action log loss over action frequency but none passed all of its separately frozen promotion requirements; none was evaluated online. A read-only residual diagnosis rejected a global probability scale as sufficient and selected one small current-observation nonlinear ablation without inspecting scaled validation. That MLP converged and improved held-out log loss, accuracy, and Brier over the linear model, but failed the unchanged calibration gate and was not run online. A frozen target-contract diagnosis then verified the recorded propensities exactly and found neither sampled-label noise, 500 more optimization steps, nor replacing one-hot targets with full propensities sufficient. The residual is localized to the flat scorer's inductive bias for an exactly reconstructible action-conditioned lexicographic rule. The selected seven-parameter shared action scorer then exhausted its optimization timebox and was worse than the MLP on both splits, with zero final-tie agreement. This exposes a deeper mismatch between compensatory pointwise scores and the known non-compensatory collector rule; no learned policy was run online. A subsequent preregistered offline factual-probe pilot found exact publication/execution alignment and held-out current-root signal for observed-shield contraction and physical HIT at supported 16- and 64-frame horizons. This is representation evidence, not a causal action-effect or gameplay-improvement result, and it admits neither history nor a learned policy. A state-only attribution then found a small incremental action-relative HIT signal at 16 frames but not 64, with worse calibration than state-only at both horizons; action-relative shield-contraction signal was robust. Because it reuses the same held-out episodes, it is selection evidence only, not independent confirmation. A further frozen support/lifecycle diagnosis found that the weak HIT increment was confined to actions equal to the reactive baseline and reversed sign on actions known to differ from it. Pre-first-HIT rows retained a negative point delta, rejecting a purely post-HIT artifact, while overprediction and clipping remained material. Eight fresh complete episodes then independently reproduced the 16-frame proper-score increment, including the preregistered nonbaseline point and pre-first-HIT boundaries, but failed calibration readiness. The predictive representation is confirmed; causal action value, history, and online policy remain unadmitted. A subsequent train-only two-parameter Platt surface repaired the frozen global calibration limits but worsened Brier against both the uncalibrated full and state-only probes. Scalar calibration is therefore rejected. A direct small current-root/action regressor trained by 16-frame physical-HIT Brier then strongly improved the frozen score and its same-architecture state-only ablation, with acceptable calibration. Its formal gate still rejected fresh confirmation because low-propensity nonbaseline and pre-first-HIT evidence was not episode-stable. An exact bounded uniform-shield-target weighting test then worsened its target Brier comparator and retained unstable low-propensity and nonbaseline action evidence, rejecting logged action measure as the primary cause. A fixed 16-root portable history then improved the point score and the overall current-action ablation, but failed episode-stable temporal, exploratory-action, pre-first-HIT, and raw-boundedness gates. It is rejected. A bounded current-root primitive-set model then improved ranking and aggregate action ablation, but not object-set Brier; exploratory and pre-first-HIT action deltas reversed, while direct sigmoid Brier saturated 65.05% of rows and underpredicted event rate. It too is rejected. Changing only its training loss to BCE-with-logits improved Brier and reduced saturated missed positives from 89 to one, but still failed the joint gate. A factual audit then showed that 99.45% of h16-positive rows hit only after the current action stopped. The target is behavior-continuation risk, not supported current-action value; the next boundary is a safe fixed-short-exposure collection contract, not a larger learner. That collection-only test is now preregistered at four policy roots and has not run. No hazard scorer or learned policy is selected.

1 Motivation and Scope

Bullet-hell control combines dense continuous geometry, discrete input, delayed input pickup, partially observed pattern state, long episodes, and sparse but unambiguous failure events. A tempting engineering strategy is to combine a large object encoder, recurrent policy, offline value learner, ensemble uncertainty estimator, active collector, and learned world model at the outset. Such a stack cannot reveal whether a failure came from factual capture, causal action linkage, representation, value learning, exploration, or online inference.

Touhou-Offline instead adopts a *capture-complete, prediction-incomplete* boundary. The adapter records what physically exists at each coherent root, while learning is responsible for regularities not encoded by the minimal observed-hazard filter. The immediate engineering target is a complete Touhou 6 Lunatic route with HIT continuation and zero Bomb. The scientific target is Lunatic NMNB. Touhou 8 is reserved as a transfer test rather than a source of training-time identifiers.

This is a working protocol, not a result paper. In particular, we do not yet claim that the observed-hazard shield prevents all HITs, that object history is sufficient to infer unknown births, that offline RL improves the behavior policy, that a fixed corpus is sufficient, that exact-prefix branching is possible, that accelerated or parallel Wine is evidence-equivalent, or that the method transfers to Touhou 8.

1.1 Frozen outcome and non-goals

The required outcome is one immutable policy that reduces physical HIT count on complete normal-speed original-Wine routes without Bomb, infrastructure failure, forbidden actor input, or widening the observed-hazard action set. Every promoted result must also report NMNB completion rate. The first result need not complete NMNB; it must establish a correctly attributed improvement.

The initial baseline does not attempt online ECL interpretation, global safety proof, exact future-birth prediction, stage or spell routing, policy updates during a run, parallel collection, counterfactual simulation, or a novelty claim. These are non-goals unless a named experiment later earns one of them.

2 Objective and Causal Unit

For a complete physical episode τ, define

$$c_t = \mathbf{1}[\text{a physical life loss is observed on transition } t], \qquad C(\tau) = \sum_{t=0}^{T-1} c_t. \tag{1}$$

Bomb is forbidden and infrastructure failures are reported separately rather than converted into reward. Clearance, graze, score, power, rank, route progress, stage identity, and source identities are not reward.

On the declared evaluation population of complete, infrastructure-valid, zero-Bomb routes, the scientific goal metric is

$$J_{\text{NMNB}}(\pi) = \Pr_{\pi}[C(\tau) = 0]. \tag{2}$$

The first optimization target is

$$J_{\text{HIT}}(\pi) = \mathbb{E}_{\pi}[C(\tau)], \qquad \gamma = 1, \tag{3}$$

with terminal value zero only at the declared physical episode terminal. The two objectives are not interchangeable: equal expected HIT count can hide very different NMNB probabilities. Nevertheless, because C is a nonnegative integer,

$$1 - J_{\text{NMNB}}(\pi) = \Pr_{\pi}[C(\tau) \geq 1] \leq \mathbb{E}_{\pi}[C(\tau)] \tag{4}$$

on that population. HIT count therefore supplies a dense initial surrogate whose claim is exactly "fewer expected HITs," not "higher NMNB probability." Incomplete or invalid trials are never removed silently: their rates and causes are a separate promotion gate. A survival or distributional objective is a separate candidate experiment and may not be introduced silently.

The raw corpus remains frame-linked, but the policy intervention is not every raw frame. A command can remain in an input lease until Wine witnesses its pickup. A derived learner row must therefore join one policy invocation to the next or to the physical episode terminal and contain the published command, its complete behavior probability, witnessed sampled/executed actions, elapsed game frames, accumulated physical HIT cost, the next decision root or terminal, and all exclusion reasons. This decision-epoch view is a causal projection of factual Wine rows, not a simulated successor or a hand-designed gameplay option.

For reward-maximizing implementations the only task reward is $r_k = -C_k$, so maximizing $\mathbb{E}[\sum_k r_k]$ is exactly minimizing J_{HIT}; a cost-form implementation may minimize C_k directly. There is no discount, stage-boundary terminal, first-HIT terminal, or shaped progress term hidden in either convention.

Post-HIT roots remain in the complete episode and in the primary HIT-count view. Separate pre-first-HIT and time-since-HIT views may measure survival-conditioned coverage or train auxiliary factual predictors, but the initial task learner may not relabel the first HIT as the episode terminal.

3 Research Questions

RQ1: portable representation. After exact decision-epoch reconstruction, do current portable physical facts predict factual behavior and future HIT risk on held-out complete episodes, and does a short physical/action history improve them without privileged identity?

RQ2: minimal policy improvement. On one frozen representation and corpus, does the smallest justified offline value learner reduce complete-route expected HIT count relative to the behavior and behavior-cloning baselines under original Wine?

RQ3: coverage and interaction. If the fixed-corpus learner fails, is the demonstrated cause inadequate action/state support, survival-conditioned occupancy, representation, or value estimation; and does one frozen-policy batch of additional Wine data resolve that named failure more efficiently than uniform collection?

RQ4: transfer. Can the episode schema, causal learner view, feature contract, objective, exporter, and evaluator remain unchanged when only the Touhou 6 environment adapter and configuration are replaced by a Touhou 8 adapter?

The source-order question is no longer an open architecture choice. TH06 permits a hazard born after one paused root to enter collision processing before the next completed root. Linking and predicting this class in Wine remains an experiment, but it cannot justify an online birth envelope or ECL interpreter.

4 System Boundary

The system has three independently testable components.

1. A Wine adapter pauses the exact process and records a coherent physical root, input delivery, behavior probability, outcomes, lifecycle, and provenance.
2. A native shield filters actions whose bounded paths overlap hazards already instantiated at that root.
3. An offline learner consumes complete generic episodes and exports a frozen scorer. The scorer may rank the shield set but may never add an action to it.

There is no mandatory online ECL interpreter, stage route, spell rule, boss rule, frame window, coordinate target, or pattern exception. Extracted source scripts may support forensic analysis of the immediate-birth boundary, but source identities are not actor inputs and their successful decoding is not a corpus-admission condition. HIT is an observed cost and does not terminate data collection; Bomb is forbidden.

5 Minimal Online Shield

At decision root t, let x_t be player state, A the 18 directional/focus actions, and O_t the instantiated bullets, lasers, and enemy collision bodies. The adapter lowers O_t to per-frame hazard primitives for a fixed horizon $H = 4$. For each candidate $a \in A$, the native kernel advances the player under the measured movement constants and bounded input-delivery paths. It returns

$$A_{\text{shield}}(x_t, O_t) = \{a \in A : d(x^a_{t+k}, O_{t,k}) > m,\ k = 1, \ldots, H\}, \tag{5}$$

where $m = 0.35$ pixels is the fixed additional collision margin and d uses the source collision geometry.

The claim is deliberately narrow: a returned action is clear of the stored projection of hazards observed at root t. The shield makes no claim about a hazard born after t. The source-order audit at TH06 commit `cc475a0bc3fef38683b0f02224c87ddba0a021d9` establishes this boundary: Player update priority 7 precedes EnemyManager priority 9 and BulletManager priority 11; enemy execution may instantiate a hazard which BulletManager can test for collision in that same update. The audited definitions are in `src/ChainPriorities.hpp`, `src/EnemyManager.cpp`, and `src/BulletManager.cpp` of that source revision. Such births are ordinary partially observed learning outcomes, not justification for a title-specific online predictor. Empty shield sets are recorded as control dead ends; they are not silently widened.

The online path therefore contains only coherent capture, observed-object kinematics, native collision filtering, frozen-policy scoring, and input publication. A retired hand-written long-horizon beam planner is excluded so that gameplay changes can be attributed to the learned policy rather than an untracked rule system.

6 Algorithm-Independent Episode Data

An episode is an immutable sequence

$$\tau_i = \{(o_t, A_t^{\text{shield}}, a_t, \mu_t, o_{t+1}, c_t, b_t)\}_{t=0}^{T_i-1}, \tag{6}$$

where o_t contains factual physical state, a_t distinguishes proposed, published, sampled, and physically witnessed execution, $\mu_t(a)$ is the complete behavior distribution over the admissible actions, c_t is the physical HIT cost, and b_t records resource deltas, lifecycle boundaries, and infrastructure exclusions. Frames and transitions are joined by immutable references and shard hashes. A complete episode has exactly one fewer transition than frame.

The learner derives decision epochs k without changing the raw episode:

$$d_k = (h_k, u_k, \mu_k, \Delta_k, C_k, h_{k+1}, e_k). \tag{7}$$

Here h_k is a versioned portable history ending at a root where the immutable policy was invoked, u_k is the published command or $\perp$ when publication did not occur, Δ_k is elapsed game frames to the next invocation or the episode terminal, C_k is the sum of factual HIT indicators over that interval, and e_k contains publication, pickup, observation-gap, and learning-exclusion evidence. A $\perp$ row is retained for accounting but is not an action-conditioned training sample. Commanded, sampled, and executed actions remain distinct. The decision rows must conserve complete-episode HIT count exactly.

The canonical raw layer retains measured player state; occupied bullet, laser, enemy, player-shot, and item records; visual and collision geometry; resource counters; input state; clocks; capture retries; and executable, native, policy, and schema provenance. Derived tensors are versioned products rather than replacements for raw shards. This permits multiple feature builders and RL algorithms to consume identical episode identities.

The data contract has four distinct planes. The authoritative Wine plane is a dense title-specific capture; the factual-transition plane binds policy and input delivery to the next observed root; the portable plane exposes only actor-admissible physical observations; and derived features, histories, labels, and models are reproducible products. A recorded field is therefore not automatically an actor input, reward, or auxiliary target.

Current control roots retain subpixel player position and hitbox, movement constants and witnessed input; raw bullet-pool tails and visual mappings; raw laser, enemy, enemy-manager, and spawn-bullet records; decoded physical enemy bodies, lasers, items, and player attacks; resource and lifecycle counters; and capture integrity. Decision evidence retains the exact four-frame observed hazard primitives, collision margin, complete admissible mask, and each action's bounded clearance and final player position. Transition evidence keeps baseline, proposed, published, commanded, sampled, and executed actions distinct and stores the complete behavior distribution, factual successor, elapsed frames, HIT/Bomb/resource outcomes, and every exclusion reason.

Data plane	Representative retained facts	Authority and use
Wine root	Player/hitbox, raw and decoded instantiated objects, attacks, items, resources, lifecycle	Physical evidence, shield input, parity replay, and later bounded portable representations
Decision and input	Shield primitives and mask, policy identities, all action witnesses, complete propensities	Temporal/action alignment, BC targets, calibration, support, and later OPE
Transition outcome	Linked successor, elapsed frames, HIT, Bomb, deltas, boundaries, exclusions	Factual dynamics/risk labels and undiscounted physical HIT cost; never counterfactual successors
Integrity and provenance	Shard hashes, executable/native/policy hashes, schemas, latency, gaps, drops, completion and cleanup	Corpus admission and attribution; never gameplay features

Table 1: Separation of recorded facts from actor-visible and derived data. Recorder richness supports auditing and future falsifiable projections; it is not permission to expose privileged identity to the policy.

The first actor projection is deliberately much smaller: player position, power, bullet/laser and admissible-action counts, current action, and for each of 18 actions its shield legality, clearance-known bit, clearance, and final position. This gives 114 scalar features. Dense object facts may support one later bounded object-set experiment, but raw addresses, stage/boss/spell/ECL or RNG identity, run or absolute-frame identity, and future outcomes are forbidden actor inputs. Score, graze, clearance, and route progress are not reward; clearance is retained only as action-conditioned observation and parity evidence.

The same admitted episode may expose three non-authoritative derived views:

1. the complete decision-epoch HIT-cost view used by the initial task;
2. a pre-first-HIT or time-since-HIT diagnostic view used only to measure survival-conditioned coverage and factual prediction;
3. a complete physical view, including post-HIT roots, used for auxiliary representation and dynamics targets.

Every view names the same episode and raw hashes. No view may invent a next state, turn an unexecuted action into a transition, or admit a diagnostic stop-on-HIT run as training evidence.

Deployable features may contain normalized player state, unordered physical object primitives, action history, declared portable resources, and the shield mask. They may not contain stage, boss, spell, ECL, RNG, memory address, run identifier, future outcome, or post-treatment facts. Unordered hazard sets motivate permutation-invariant encoders such as Deep Sets [4], but this complexity is admitted only after a transparent vector baseline passes.

7 Minimal Learning Architecture

The strongest simple baseline has exactly three independently changeable parts:

1. a deterministic builder that converts complete factual episodes into audited decision rows and whole-episode splits;
2. a transparent current-observation behavior-cloning scorer,

$$\mathcal{L}_{\mathrm{BC}}(\theta) = -\sum_{i,k} \log \pi_\theta(u_k \mid h_k, A_k^{\mathrm{shield}}); \tag{8}$$

3. an immutable exporter that reproduces the offline action distribution bit-for-bit, ranks only observed-shield-admissible actions, and is evaluated on complete original-Wine episodes.

Action frequency and the existing reactive policy are non-fitted controls. BC is a plumbing and representation test, not a policy-improvement claim. The first known constraint is that the current online policy API exposes scalar player facts, object counts, and shield summaries but not the physical object history contemplated by RQ1. The baseline intentionally uses that smaller surface first. An API expansion is earned only after an offline held-out experiment shows that a precisely specified additional portable history improves a factual target and after offline/online feature parity is testable.

8 Candidate Ladder and Admission Gates

Only one rung may change at a time. Every rung freezes its corpus inventory, whole-episode split, feature schema, target schema, seeds, acceptance statistic, and artifact hashes before fitting.

This ladder controls attribution, not model sophistication for its own sake. After a boundary is measured, the next rung should be the smallest representative algorithm that directly addresses it; the study need not exhaust every weaker model family first. Conversely, an unfocused architecture or hyperparameter sweep cannot substitute for a named mechanism and a frozen decision rule. A formally negative prototype may therefore select a modified member of the same algorithm family when its diagnostics select a falsifiable mechanism, as L2f selected the subsequently negative L2g action-measure test.

L0: causal data audit (completed). Build and test decision-epoch rows. Admission requires exact raw linkage, command/sample/execution separation, propensity normalization, interval and episode HIT conservation, exclusion replay, forbidden-feature rejection, and two materially different feature builders reading the same episode identities. No model result is meaningful before L0 passes. The implementation and E0 replay evidence described in E2 satisfy this gate.

L1: transparent supervised baseline (completed negative). After L0, collect the initial training inventory serially, with unchanged retail clock/update semantics, coherent capture suspension, the frozen declared 20% uniform observed-shield mixture, and complete episodes; this is non-adaptive baseline collection, not L5. Freeze the episode count and split before fitting. Report per-admissible-mask action frequency, the reactive behavior baseline, and current-observation BC. BC passes the learnability test only if its held-out complete-episode negative log likelihood improves over action frequency with a preregistered episode-bootstrap interval and its calibration does not regress beyond a frozen tolerance. Accuracy alone is insufficient. E3 passed the likelihood condition but failed calibration, so L1 emitted no online candidate and does not admit L2 without a new transparent preregistration. Read-only replay subsequently bounded the first failure at optimization rather than capture: the portable roots reconstruct the recorded reactive choice exactly, train and validation show the same under-confidence, and the frozen train gradient is not stationary. L1b therefore changes only the train-only stopping rule and corrects the ECE implementation; it does not change data, split, representation, target, model class, regularization, or comparator. L1b exhausted its frozen update timebox before convergence and is inconclusive, so it still does not admit L2. L1c is the narrow continuation: it restarts the same fit from zero and changes only the maximum update ceiling from 2,000 to 10,000, retaining the same train-only 0.01 relative-gradient early stop and evaluating validation exactly once after stopping. No optimizer, hyperparameter, representation, comparator, or gate changes. L1c reached the gradient gate and failed calibration, rejecting this linear current-observation candidate under the frozen joint gate. A read-only residual decomposition may choose one subsequent representation ablation; L2 is not yet admitted. That diagnosis is frozen before execution: it may fit one positive global logit scale on training episodes only, but it does not evaluate the scaled validation distribution or fit a deployable model. If train calibration is repaired, it selects a scalar-calibration experiment; otherwise exact feature-only reactive reconstruction selects one small current-observation nonlinear model. History is not a candidate for this behavior target because the declared collector distribution is a current-root reactive choice plus an independent uniform draw.

L2: factual action and risk probes. First fit current-observation action-conditioned probes for executed $\Delta x/\Delta y$, observed-shield-set collapse, and physical HIT within fixed horizons. These diagnose command alignment and whether the captured current root contains learnable risk signal; they do not alter reward or make a gameplay claim. Add one short, fixed physical/action history only if a current-observation probe fails its separately frozen proper-score gate and the failure is attributable to temporal ambiguity. History is retained only if it then improves the same held-out target with an episode-level interval. Deep Sets [4] are considered only if a transparent bounded vector summary fails; a recurrent encoder is considered only if history has already shown value. Auxiliary heads remain separate from reward.

L3: one offline value method. Only after L1–L2 establish a usable representation may BC be compared with one expected-HIT value learner on the identical data. IQL is the first candidate because its critic avoids directly evaluating unseen actions and its actor remains advantage-weighted behavior cloning [2]. CQL is not run in parallel; it is admitted only if the IQL experiment identifies unsupported-action overestimation that its conservatism is designed to address [3]. Offline metrics can reject a candidate but cannot promote it without complete Wine evaluation.

L4: objective alternatives. A bounded survival or distributional HIT model is admitted only when the corpus contains enough late no-HIT prefixes or zero-HIT episodes to avoid a degenerate all-failure target, or when policies with similar expected HIT count show a measured difference in NMNB probability. Until then, first-HIT survival is a diagnostic, not the task terminal or deployment score.

L5: interaction and uncertainty. Frozen-policy batch collection is admitted only after L3 demonstrates either policy improvement or a measured support failure. The first additional batch comparison repeats the declared uniform-shield mixture at a fixed interaction budget. Ensemble uncertainty, active collection, OPE, or baseline bootstrapping may each be tested later against one named support or calibration failure. A positive action-probability floor does not by itself establish state coverage or reliable long-horizon OPE.

L6: learned dynamics and exact-prefix branching. Learned birth models or risk-aware MPC are considered only if model-free value learning fails while held-out future-hazard prediction succeeds. Predictions remain scores and never extend the shield certificate. Real-Wine branching is considered only after a serial same-seed, same-policy, exact-action-prefix gate reaches a target root with identical factual and input-delivery digests. The negative parallel gate does not establish this ability. Every training branch must still be a complete admitted Wine episode.

Decision Transformer [1], large recurrent world models, simultaneous IQL/CQL sweeps, and the proposed SABER-RL composite (survival critic, recurrent belief, ensemble uncertainty, active collection, and Wine branching) are not active candidates. SABER-RL is decomposed across L2 and L4–L6; neither the name nor the combination is a novelty claim. These methods add independent failure modes before the baseline has exposed a constraint they solve.

8.1 Algorithm and policy record

Table 2: Algorithm record at this checkpoint. "Not run" means no fit and no online learned candidate, not an unreported negative result.

Policy or method	Status	Evidence or admission condition
Reactive scorer	Executed control	E0 completed one route with 108 HITs; this is infrastructure evidence, not a learned-policy result.
20% uniform-shield mixture	Executed collection	Twelve new serial Stage 4 episodes completed on attempt 1 with 139 factual HITs; this is a training inventory, not policy-improvement evidence.
Action frequency	Evaluated comparator	Validation NLL 2.286394 and ECE 0.008749 on the frozen four-episode split.
Current-observation BC	Negative held-out gate	Validation NLL 2.014976 beat action frequency, but ECE 0.124251 exceeded the frozen 0.028749 bound; no Wine canary ran.
Current-observation BC convergence follow-up	Inconclusive	Validation NLL improved to 1.720755 and ECE to 0.055148, but 2,000 updates ended at gradient ratio 0.0196 rather than the frozen 0.01 threshold; no canary ran.
Current-observation BC timebox extension	Negative held-out gate	The identical fit converged at update 4,631 and validation NLL improved to 1.691664, but ECE 0.059713 exceeded the unchanged 0.028732 limit; no canary ran.
L1c residual decomposition	Completed	Train-only scaling remained miscalibrated; current features reconstructed the collector exactly while the linear model reproduced only 66–67% of its reactive choices, selecting one small current-observation MLP.
Small current-observation MLP	Negative held-out gate	The fixed 114–32–18 model converged and improved validation NLL to 1.568672, but ECE 0.076007 exceeded the unchanged 0.028732 limit; no canary ran.
L1d target-contract diagnosis	Completed	All propensities were exact; paired train-only continuations rejected sampled-label noise, stopping, and discarded full propensities as sufficient explanations, selecting one structured current-observation scorer.

Continued on next page

Policy or method	Status	Evidence or admission condition
Shared action-conditioned BC	Inconclusive	Exhausted 10,000 updates at gradient ratio 0.011684; validation KL 0.986686 and reactive agreement 0.655371 were worse than L1d, with zero final-tie agreement.
Current-observation dynamics and factual risk probes	Positive diagnostic	Published and first executed actions agreed on all 94,162 validation roots; motion MSE ratio was 0.00732. Shield contraction passed at all four horizons and HIT passed at supported 16/64-frame horizons; no history or policy fit.
Incremental action factual-risk ablation	Positive, weak attribution	Full beat state-only HIT Brier at 16 frames by a barely negative episode interval but not at 64; full calibration was worse. Shield contraction improved robustly at all horizons. Reused heldout, not confirmation.
Support/lifecycle decomposition	Completed diagnostic	The old h16 increment was baseline-equal and reversed on known-nonbaseline rows; pre-first-HIT remained favorable. No fit or policy.
Fresh h16 factual-risk confirmation	Positive signal, not policy-ready	Eight new complete episodes reproduced the full-versus-state Brier increment in all episode directions and passed weak nonbaseline and pre-first-HIT point gates, but failed calibration readiness.
Train-only Platt calibration	Negative probability-surface gate	The two-parameter monotone fit converged and passed calibration limits, but worsened h16 Brier versus both uncalibrated full and state-only and reversed both diagnostic point deltas.
Direct action-conditioned h16 hazard	Negative formal gate; positive algorithm signal	Direct full Brier was 0.00480785 versus 0.00551139 for frozen L2 and 0.00500959 for same-architecture state-only, with both episode-bootstrap intervals wholly favorable. Calibration passed, but nonbaseline and pre-first-HIT gates did not; no fresh confirmation ran.
Uniform-shield-target weighted h16 hazard	Negative action-measure gate	Exact weights passed, but weighted full Brier 0.00454350 was worse than frozen L2f at 0.00449955; overall, low-propensity, nonbaseline, pre-first-HIT, and candidate-bound gates failed. No fresh confirmation ran.
Fixed short portable-history h16 hazard	Negative representation gate	Exact 16-root history improved point Brier, but temporal, exploratory-action, pre-first-HIT, and raw-boundedness gates failed. No fresh confirmation ran.
Observed-primitive-set h16 hazard	Negative representation/surface gate	Object-full Brier 0.00473906 was worse than scalar-only 0.00468783; low-propensity and pre-first-HIT action deltas reversed, and 65.05% of full probabilities saturated below 10^{-7}. No fresh confirmation ran.

Continued on next page

Policy or method	Status	Evidence or admission condition
Log-score primitive-set h16 hazard	Negative loss/surface gate; positive mechanism diagnosis	Object-full Brier improved from frozen L2i 0.00473906 to 0.00464423, but only five episode directions favored it and scalar-only remained better at 0.00463179. Exploratory-action, lifecycle, calibration, and raw-saturation gates failed; saturated missed positives nevertheless fell from 89 to one.
Current-action/h16 target audit	Decisive collection-boundary diagnosis	The initial action lasted a median one frame, while 1,079/1,085 h16 positives first hit after it ended; the low-propensity figure was 157/158. The factual target is behavior-continuation risk, not supported action value.
Four-root randomized action exposure	Passed collection boundary after bounded audit erratum	Two serial complete Stage 4 pilots finished with 18 and 12 HITs and zero Bomb. The immutable as-run result rejected only because the checker omitted five preregistered control-dead-end interruptions. A separately hashed read-only erratum changed no facts or thresholds and passed all seven gates: 12,936 eligible groups, 93.68% complete intended execution, 0.217% dead ends, and 92 h16-positive starts. No model or learned policy ran.
Twelve-root randomized intention protocol	Preregistered, not run	Read-only sizing found zero strict h4 execution positives and only four accepted h4 intention-to-treat positives. Upper-bound event counts at 4/8/12/16 roots were 4/17/21/22, selecting the shortest near-saturated horizon at 12. Two serial complete Stage 4 pilot-train episodes test the exact h12 group target; no model runs.
IQL	Not run	First L3 value candidate after representation learnability.
CQL	Not run	Admitted only after measured IQL unsupported-action overestimation.
Survival/distributional value	Not run	Admitted only by L4 coverage or objective evidence.
Recurrence, ensembles, active collection, learned MPC, branching	Not run	Each requires its distinct L2, L5, or L6 gate; none is active architecture.
SABER-RL composite	Not admitted	Its survival, recurrence, uncertainty, collection, and branching claims must each pass their separate rung before a combined-method experiment exists.

9 Experimental Protocol

Every result must name the Git commit, executable hash, policy hash, schema, full command, complete episode IDs, and artifact paths. Splits are by complete episode. Negative results are retained. Infrastructure failures are reported separately and never counted as gameplay improvements. Before a fit, the experiment must additionally freeze its corpus inventory, split-group hashes, decision-row schema, feature and target schemas, seeds, comparator, acceptance statistic, and stopping rule. A post-hoc metric may be reported but cannot promote the candidate.

ID	Decisive question	Status
E0	Does the reset runner complete Practice 4–6 and one full route?	Completed
E1	Can a post-root birth hit before the next controllable root?	In progress
E2	Does the causal learner view preserve actions, propensities, and HITs?	Completed
E3	Do portable supervised baselines generalize by episode?	Negative result
E4	Does one offline value learner reduce complete-route HIT cost?	Not run
E5	Does shared-host parallel collection preserve exact facts?	Negative result
E6	Does a new frozen Wine batch resolve a measured coverage failure?	Not run
E7	Which surfaces change under a Touhou 8 adapter?	Not run

Table 3: Preregistered experiment ledger. Status changes require immutable artifacts.

9.1 E0: infrastructure baseline

Run natural Lunatic Practice Stages 4, 5, and 6 and then one natural six-stage route on original Wine with unchanged retail update semantics, coherent capture suspension, and the frozen baseline policy. HIT continuation and zero Bomb are mandatory. Accept only if every declared run completes, frames equal transitions plus one, no record is dropped, player successors and stored shield decisions replay exactly, every HIT has linked before/action/after evidence, and cleanup leaves no worker process. HIT count is measured, not gated.

At solver commit `af9900524520b72934a4c55e2f44118f88094633`, all three bounded Practice runs completed under isolated four-CPU workers after the atomic-input cleanup. Table 4 reports their complete audits. Every declared frame and transition loaded, every Bomb/callback/integrity/scope count was zero, all checked player successors were bit-exact, and every stored dense shield sample reproduced with no unsafe or conservative divergence. The higher Practice observation-gap rates motivated eight CPUs for the canonical route and future evidence collection; gap transitions remain explicit and are excluded rather than fabricated.

Stage	Frames	Transitions	HIT	Gap %	Successors	Shield
4	25,564	25,563	18	0.536	24,344	58
5	23,129	23,128	23	0.968	19,503	64
6	30,981	30,980	34	0.507	28,763	64

Table 4: Canonical E0 Practice evidence. Successor and shield columns are checked counts; every mismatch/divergence count is zero.

The Stage 4–6 run IDs are, respectively, `20260821T080408Z-471737400`, `20260821T081924Z-372154700`, and `20260821T080532Z-100425500`; their audit SHA-256 digests are `c4178c3c9e5b7d88f7a950d8fe18403630a8524320ae72dc0d0ce97c0a26bb2d`, `2677f2b9689c42c5483a2e278d077a6b721623430d89f6c39a41bc92b3c34ebd`, and `e34bec2844dc7ce92f572326dd262a1e7f718d6798bc9a6c9dd7124692308c17`.

The natural-RNG full route then reached Ending with run ID `20260821T083303Z-499458600`: 135,561 frames, 135,560 transitions, 108 physical HITs, zero Bomb, zero dropped record, and zero capture, corpus, policy, trace, or other infrastructure failure. All six Stages were present. The audit classified all 108 HITs as observed-shield-empty, reproduced 64 dense shield samples without divergence, and matched 126,939 eligible player successors bit-for-bit. Only 28 links were excluded for observation gaps (0.0207%); capture and solve p99 were 8.044 and 0.918 ms respectively. The independent baseline verifier passed every declared check. The immutable SHA-256 digests for report, audit, run metadata, and manifest are respectively `f2577d31ca5a51108911b4d60a957a7e`

2e891edda5ba337fbca1f70386e37686, 8d13d3196261f4f8710d8187cdec898e364b67f024696a8c ca1991aff467baa9, 9f9fbdd75dbe89651c789dc502dab6a50323a93ca48805172053830807 9e3396, and 8423bae1ef93decd50605a5e47900ee0ded3a6096d6067fb4c0a8eca944ec1d8. E0 is therefore complete; the 108-HIT value is a baseline measurement, not a policy-quality claim.

9.2 E1: immediate-birth boundary

Use source update order and adjacent physical roots around every HIT and newly observed overlapping object. Source order already proves that post-root birth and same-update collision are possible. The remaining experiment links this class to adjacent Wine roots and measures whether short physical history predicts its action-relative cost. No source/ECL envelope is added to the online shield; pattern-specific exceptions are forbidden.

9.3 E2: causal learner view

The existing generic loader already admits complete E0 episodes and enforces frame/transition linkage. E2 is not complete until it also removes optional source/ECL material, derives decision epochs from policy invocations and input leases, reproduces proposed/published/sampled/executed action sequences, conserves interval and episode HIT totals, validates complete propensities and exclusions, and lets two different feature builders consume the same raw episode identities. A feature-parity fixture must prove that the online exporter receives exactly the versioned values produced by offline replay. E2 changes no policy and collects no new gameplay evidence. At commit 2818861f4079bebfbd8443638ed0cb34236bd5e0, the decision builder, two-path offline/online feature-parity fixture, masked linear BC fixture, and immutable export/reload tests passed as part of 176 repository tests. Read-only replay of canonical E0 Practice Stage 4 produced 24,385 decision epochs, retained 24,249 as action-conditioned learner rows, and conserved all 18 physical HITs; 7 HITs remained explicitly attached to excluded intervals. The canonical complete route produced 126,468 decision epochs, retained 126,424, and conserved all 108 HITs; one HIT remained in an excluded interval. Raw-frame and compact-context actions and propensities are checked exactly, forbidden source facts are absent from the portable root, and offline and online feature vectors are bit-exact. E2 is therefore complete.

9.4 E3: supervised representation baselines

After E2, collect the initial inventory serially with unchanged retail clock/update semantics, coherent capture suspension, and the frozen 20% uniform observed-shield mixture; its episode count, seeds, and stopping rule are declared before the first run. Then freeze its whole-episode train/validation split and compare the per-admissible-mask action-frequency control with current-observation BC. Only if the preregistered held-out negative-log-likelihood and calibration gate passes may one short fixed history be added. The same split may then support simple executed-action factual probes for HIT within fixed horizons and future observed-shield collapse. These probes test information content; they do not define reward, infer unexecuted successors, or constitute policy improvement.

The first E3 inventory is frozen in experiments/l1-stage4-bc-v1.json. It contains exactly 12 complete natural-RNG Lunatic Practice Stage 4 episodes under the 20% uniform observed-shield mixture, with independent preregistered policy RNG streams. Episode indices 2, 5, 8, and 11 are validation; the remaining eight are training. Collection uses one isolated eight-CPU Wine worker at a time because E5 did not admit parallel Wine. A physical HIT never causes retry. Bomb, incomplete storage, infrastructure failure, semantic/parity failure, capture p99 above 40 ms, solve p99 above 16.7 ms, observation-gap rate above 0.5%, or nonzero stale-retry rate rejects that attempt; each fixed policy slot permits at most three infrastructure attempts, after which the experiment stops incomplete without fitting.

The current-observation masked linear softmax fit is frozen at 100 full-batch epochs, learning rate 0.05, $L_2 = 10^{-4}$, seed 0, 2,000 episode-bootstrap draws, and at most 400,000 rows per split. BC passes only when the upper end of the 95% episode-bootstrap interval for validation NLL minus the per-mask Laplace action-frequency NLL is below zero and ten-bin ECE is no more than 0.02 worse than the comparator. No auxiliary target is active in L1. A passing artifact earns one complete natural-RNG Stage 4 online integration canary; that canary is excluded from training and makes no HIT-reduction claim.

At solver commit `559572a8f3699c06eb41080e6061e579a1156c33`, all 12 slots completed and passed admission on attempt 1 with no Bomb, rejection, retry, stale input, drop, semantic/parity failure, or leftover Wine worker. The inventory contains 296,667 authoritative roots, 296,655 transitions, 794 events, 287,545 decision epochs, 287,493 eligible rows, 139 conserved HITs, and 643 observed-shield control dead ends. All HITs remained in eligible decision intervals. Capture p99 ranged from 8.018 to 8.421 ms and solve p99 from 0.764 to 0.897 ms.

The manifest-declared corpus occupies 4,583,135,878 compressed bytes for 36,491,160,547 canonical uncompressed bytes, a 7.962-fold ratio. A post-hoc, non-promotional codec diagnostic streamed one fixed 157,969,806-byte frame shard: current gzip-3 used 17,178,795 bytes, gzip-9 14,779,903, zstd-3 7,327,695, and zstd-9 5,871,707. No shard was changed. L1 therefore retains the existing format; a future codec migration requires a versioned manifest and content-equivalence audit, while a compact derived decision cache is admitted only if repeated fitting is measured to be decode-bound.

The fit used 193,331 training rows and 94,162 validation rows. Validation BC NLL was 2.014976 versus 2.286394 for action frequency; the 2,000-draw episode-bootstrap interval for their difference was $[-0.287520, -0.254559]$, so the likelihood criterion passed. Validation ten-bin ECE was 0.124251, while the frozen comparator-plus-tolerance bound was $0.008749 + 0.02 = 0.028749$. Calibration therefore failed and the joint gate rejected policy `linear-behavior-clone-v1:cdb68b242791738d`. The result artifact decision is `stop-l1-bc-learnability`; no Wine canary, HIT comparison, or NMNB claim exists. The collection ledger, model, and result SHA-256 digests are respectively `614e44b1eab047041337a5ebcee01af3ef672d01a265d856d0dd58c530dc4c87`, `7b47fee4c261eb8dbde77e3e13391a49cc5771d27e9c4d3d546f6abb695cd245`, and `75810a82c9bf3ec0a1063b0d53e6814243afc08b3319268800a2eaff5ef5bae3`.

The result establishes learnable current-observation signal relative to action frequency but not an admissible behavior distribution. A read-only reliability decomposition may diagnose the frozen artifact; any refit, temperature calibration, optimizer change, or representation change is a new preregistered experiment. E4 and online learned-policy evaluation remain blocked.

That read-only decomposition reproduced the BC metrics and found systematically low confidence rather than a validation-only failure. Validation accuracy was 0.461290 while mean top confidence was 0.337672; train showed 0.468140 versus 0.341038. Each validation episode had ECE between 0.1151 and 0.1362, only 0.808% of validation rows used an unseen training shield mask, and the largest feature-mean displacement was 0.214 training SD. The frozen train gradient L2 norm was 0.197788, 19.1% of its initial 1.037370, while the train-NLL derivative along a uniform logit-scale increase was -0.271893. Thus the 100 full-batch updates did not reach a stationary train solution.

The permitted portable facts independently reconstructed the recorded reactive baseline on all 94,162 validation rows, excluding missing capture and the future-birth boundary as explanations for this action-imitation result. The linear BC reproduced that baseline on only 56.19% of rows. This may still expose linear expressivity after convergence: 58.3% of rows tied at maximum clearance and 43.4% had all clearances infinite, so the controller frequently used later tie breakers including nonlinear boundary reserve. That hypothesis is not admitted until optimization is resolved.

The decomposition also found that floating ECE edges failed to assign exact confidence 0.6 to any bin. Integer equal-width binning changes the frozen action-frequency ECE from 0.008749 to 0.008732 and leaves BC ECE at 0.124251; the corrected bound is 0.028732, so the historical L1 decision remains negative. The original result is not rewritten.

The separately frozen L1b protocol is `experiments/l1b-stage4-bc-convergence-v1.json`. It reuses the exact 12-episode inventory and eight/four split, initializes the identical masked linear softmax at zero, retains learning rate 0.05 and $L_2 = 10^{-4}$, and uses the corrected versioned ECE. After at least 100 updates, training stops when gradient L2 is at most 0.01 of its initialization value, or after 2,000 updates. The latter outcome is inconclusive and cannot be attributed to representation. Validation is evaluated exactly once after the train-only stop. The likelihood and comparator-plus-0.02 calibration conditions are unchanged. L1b collects no data and runs no Wine canary; a joint pass only admits a separately executed Stage 4 integration canary.

L1b ran at solver commit `7285ee76fe36eda1470844b3635eaddd64292d23`. It exhausted all 2,000 updates with final train gradient L2 0.020297, or 0.019566 of the initial 1.037370, and therefore missed the required 0.01 ratio. The optimization gate failed and the preregistered result is `inconclusive-l1b-optimization-not-converged`; this is not evidence that the linear representation failed.

Optimization nevertheless moved every descriptive held-out metric in the expected direction. Validation NLL fell from the L1 value 2.014976 to 1.720755, Brier from 0.728516 to 0.647767, ECE from 0.124251 to 0.055148, and accuracy rose from 0.461290 to 0.528111. The corrected action-frequency NLL and ECE were 2.286394 and 0.008732; the episode-bootstrap NLL-difference interval was $[-0.578257, -0.552904]$. The proper-score gate passed, but ECE remained above the corrected 0.028732 bound, so calibration also failed. Train NLL and ECE were 1.707485 and 0.054720, close to validation. No online canary ran and no HIT or NMNB claim exists. The plan, model, and result SHA-256 digests are respectively `d972fbda8b26cf37fe69438fc81423714c92f652652ba1667e5c27830bcdb592`, `22254afd55d97a740889e76c44617fd408d3354566a550abc3f74feb9b497ae5`, and `0d814a6f20036175e16a507256e79c3195ce4b48193b73c3fc51b7cc33d1d336`. Any convergence-suitable optimizer or larger timebox would be another frozen L1 experiment; validation may not select it.

The L1c protocol is frozen in `experiments/l1c-stage4-bc-timebox-v1.json`. It reuses the exact L1 corpus, whole-episode split, code, from-zero initialization, features, target, linear model, full-batch optimizer, learning rate, regularization, seed, comparators, and acceptance gates. Its single experimental change is a maximum of 10,000 rather than 2,000 updates. After at least 100 updates it still stops immediately when train gradient L2 is at most 0.01 of the same initial value; validation is evaluated once only after that train-only stop. The ceiling is five times L1b's optimization budget because L1b ended less than a factor of two above the frozen gradient threshold. This is a resource bound, not a validation-selected update. L1c collects no Wine data and runs no canary. If it converges but fails calibration, the transparent linear current-observation representation is rejected by the unchanged gate. If it still does not converge, the next smallest experiment is a separately frozen deterministic convergence-suitable optimizer, not history or a larger model. L1c ran at solver commit `171d92b95c55331ab23bec73226b6b81de6f64cf`. The train-only stop fired at update 4,631 with final gradient L2 0.010372, or 0.009998 of the initial 1.037370. The optimization gate therefore passed. Validation NLL was 1.691664 and the episode-bootstrap BC-minus-frequency interval was $[-0.607988, -0.581455]$, so the proper-score gate also passed. Validation ECE was 0.059713 against the unchanged comparator-plus-tolerance limit $0.008732+0.02 = 0.028732$; calibration and the joint gate failed. Validation accuracy and Brier were 0.543234 and 0.634249. Continued optimization from L1b improved NLL, Brier, and accuracy but worsened ECE by 0.004565. The preregistered decision is `stop-l1c-linear-current-observation`; policy `linear-behavior-clone-v1:368d286c449c3d7d` was not run online.

The immutable plan, model, and result SHA-256 digests are respectively `f1eef124b1717cac7f582a50964e793ed98a75e34ea41930e66a6f734d4cb8aa`, `b7e350ff86103ee803a6c2a856f5509b1a29b5080c02e78e5825de9a9cca16b9`, and `fac6fd06cd57f93ed276a6c5afa5a3c6c8da9bf8dd42935fac6b876ffe3dc1a1`. L1c rules out insufficient optimization time as a sufficient explanation for the calibration failure. It rejects the current feature/model/exported- probability combination under the declared gate, not BC in general or every model using current physical observations. It also does not distinguish nonlinear current-observation expressivity from missing temporal information. One read-only residual decomposition and then one separately frozen representation ablation are admissible; continued optimization of the same fit, an online canary, combined history/architecture changes, and IQL are not.

The residual protocol is frozen in `experiments/l1c-stage4-residual-diagnosis-v1.json`. It binds the L1c model and result, replays the same eight/four whole-episode split, and first reproduces all frozen metrics. A one-dimensional positive logit scale is fit by training NLL only. Branch selection uses its training NLL and ECE plus an exact reconstruction of the reactive baseline from the 114 current features; the scaled validation distribution remains unread. A successful train-only scale selects one separately preregistered scalar-calibration validation. Otherwise, exact current-feature reconstruction selects one small nonlinear current-observation model. Failure of the reconstruction invariant stops the ladder rather than silently selecting history.

The diagnosis ran at solver commit `a4e8232fbae5e3c4df9ecb8d7a787c5138d54e32`. The positive train-NLL optimum was logit scale 1.032470 (temperature 0.968551) and changed no argmax. It improved train NLL only from 1.679447 to 1.678613 and train ECE from 0.058741 to 0.046387, still above the frozen 0.028732 limit. The scaled validation distribution was not evaluated. Current features reconstructed the reactive baseline on every one of 193,331 train and 94,162 validation rows, ruling out temporal ambiguity for the collector target; its remaining 20% choice is an independent declared uniform draw. The linear model agreed with the reactive action on only 67.05% of train and 66.40% of validation rows. The exact collector oracle had train/validation NLL 1.004524/1.006665 and validation ECE 0.000756, so this gap is not irreducible exploration noise. The preregistered selection is therefore one small current-observation MLP, with temperature, history, auxiliary targets, and value learning excluded.

The immutable plan and diagnosis SHA-256 digests are respectively `a05cd1f70f2e689eb5018cf3393dd30c3681006ae7cf84b834cc5193b8e3bbab` and `e3c25c844c9ec6fdd47388fb3a9690375dcc77c2c7963d30ebfa180eaccce8ce`.

The selected L1d representation ablation is frozen before fitting in `experiments/l1d-stage4-bc-mlp-v1.json`. It reuses the immutable 12 episodes, eight/four whole-episode split, 114 current-observation features, shield mask, published executed-action target, learning rate 0.05, L2 10^{-4}, full-batch gradient descent, seed 0, 10,000-update ceiling, and the same train-only 0.01 relative-gradient stop. Its sole representational change is a fixed 114–32–18 network with one ReLU hidden layer. All-zero weights would leave the hidden units symmetric, so the one necessary companion change is fixed-seed He-normal weights with zero biases; initialization, width, and depth are not swept. There is no history, temperature, auxiliary target, privileged collector action, value loss, or new Wine data.

L1d's primary proper-score test is deliberately harder than the original L1 frequency comparison: the upper endpoint of a 2,000-draw complete-episode bootstrap interval for MLP-minus-frozen-L1c validation NLL must be below zero. It must also converge under the unchanged train-only criterion and keep validation ECE no greater than action-frequency ECE plus 0.02. Improvement over action frequency remains report-only and cannot replace direct evidence against the converged linear model.

Validation is evaluated once after the train-only stop. Non-convergence is inconclusive; convergence followed by a failed direct-NLL or calibration gate rejects this small current-observation MLP. This offline fit runs no Wine canary and makes no HIT, value-learning, or NMNB claim. Only the joint supervised gate may admit a separately preregistered Stage 4 integration canary.

L1d ran at solver commit `c96845952572f41f4c263706fe76548b6927df7e`. Its train-only stop fired at update 3,121 with final gradient L2 0.020971, or 0.009996 of its fixed-seed initial value 2.098068, so optimization passed. Validation NLL was 1.568672, compared with 1.691664 for frozen converged L1c; the complete-episode MLP-minus-L1c bootstrap interval was $[-0.127334, -0.115239]$, so the direct representation gate passed. Validation accuracy and Brier also improved from 0.543234 and 0.634249 to 0.610076 and 0.572805. The report-only MLP-minus-frequency NLL interval was $[-0.734694, -0.700380]$.

Calibration nevertheless failed. Validation ECE was 0.076007 against the unchanged limit $0.008732 + 0.02 = 0.028732$; train ECE was similarly high at 0.080346. A read-only post-result reliability check found mean validation top confidence 0.549501 versus accuracy 0.610076. Most 0.2–0.7 confidence bins were under-confident, while the 0.8–0.9 bins were over-confident; no probability transform was fit or evaluated. The immutable decision is `stop-l1d-small-current-observation-mlp`. The plan, model, and result SHA-256 digests are respectively `ef1ae999fe75d434cd60f9126d50a2d7e89210f88e925626174e117a0dc7e8c5`, `3f8d54f2574e4830c0b2cc81c516d70bcc5637967600a8df8941860a8ddaa286`, and `aaf62b7e6a696c4d72e7f61a8b68e772d45e6f320b23c7e963b84b15d7657bf2`. Thus the nonlinear current-observation hypothesis receives partial support on proper score, accuracy, and Brier, but this exact exported probability model fails the joint supervised gate. It runs no Wine canary and admits neither history nor value learning. Any calibration diagnosis or further ablation must be separately frozen; this result cannot be repaired post hoc.

The root-cause protocol is frozen in `experiments/l1d-stage4-target-diagnosis-v1.json`. It first audits every eligible row against the declared 0.8 reactive plus 0.2 uniform-shield distribution and measures frozen L1d against both sampled actions and the recorded full distribution. It then starts two non-deployable branches from the exact frozen L1d state: 500 additional full-batch train-only updates with the historical one-hot published-action objective, and 500 with cross-entropy to the recorded full propensities. Both retain learning rate 0.05 and L2 10^{-4} and report fixed checkpoints 0, 100, 250, and 500. Branch parameters are never serialized and neither branch is evaluated on validation.

The diagnosis attributes a recorder/dataset defect only if a recorded mixture differs by more than 10^{-15}. Otherwise a final soft-target train KL advantage of at least 0.005 nat selects one separately preregistered full-propensity target correction. Failing that, a hard-target KL reduction of at least 0.01 nat selects one stopping correction. If neither occurs while a large distribution and reactive-action residual remains, the next candidate is one structured current-observation scorer, not history. This diagnosis runs no Wine, fits no deployable model, does not revise the negative L1d result, and cannot admit value learning.

The diagnosis ran at solver commit `b12efebd995832fd68669af163255419cd517dc7`. All 193,331 train and 94,162 validation rows matched the declared behavior distribution with maximum absolute error exactly zero. Frozen L1d validation $\mathrm{KL}(\mu\|\pi)$ was 0.563604 and exact-distribution top-label calibration error was 0.077039, close to sampled-action ECE 0.076007; sampling noise is therefore not a sufficient explanation. The current features reconstructed the reactive action on every row, whereas L1d agreed on only 0.754633 of train and 0.748550 of validation rows. On validation, agreement was 0.801009 when boundary reserve was decisive and 0.725546 when clearance was decisive, but only 0.003086 for focused and 0.039474 for lexical final tie breaks.

After 500 additional hard-target train updates, train $\mathrm{KL}(\mu\|\pi)$ fell from 0.552480 to 0.542529, a 0.009951 reduction just below the frozen 0.01 materiality threshold. The paired full-propensity branch ended at 0.541191, only 0.001338 nat better than the hard branch and below the frozen 0.005 threshold. Thus the result does not claim that further optimization or exact soft labels have zero effect; it rejects either as sufficient for the large residual under the preregistered bounds. The selected attribution is a flat-model inductive-bias mismatch for the known action-conditioned lexicographic relation. History remains unsupported. The experiment-plan and result SHA-256 digests are `eabe1743d40bc1cb8d5ab949826658ac46e483ef564ce5e922a4f84739f05738` and `6aaee7631249dc30f0ac60fa76a3e72a2c42ab5bda580340ca6b14722b056e4f`, respectively.

The diagnosis also exposes two prospective protocol corrections without changing any historical decision. First, reducing a captured full behavior distribution to one sampled action discards exact supervision, although its measured effect here was only 0.001338 nat. Second, top-label ECE relative to a diffuse action-frequency predictor is not a proper distributional comparison: such a predictor can have small ECE despite poor likelihood. A future behavior-model gate should compare episode-bootstrap cross-entropy or $\mathrm{KL}(\mu\|\pi)$ directly against the recorded full distribution. L1d remains negative because it is also poor under that exact target; this is not a post-hoc revision of its frozen gate.

The selected L1e test is frozen in `experiments/l1e-stage4-bc-shared-action-v1.json`. It retains the same 12 episodes, eight/four split, published executed-action target, observed-shield mask, learning rate 0.05, L2 10^{-4}, all-zero initialization, 10,000-update ceiling, and train-only 0.01 relative-gradient stop. Its only learner change is one seven-parameter linear score shared across all 18 actions. For each action it reads clearance unknownness and value, boundary reserve derived from the already captured final coordinates, equality to the current action, stationary and focused action attributes, and fixed lexical action-name rank. It receives neither the recorded reactive winner nor any temporal, source, or future fact. The shared score is masked by the unchanged shield and converted to a categorical distribution.

The primary gate is now the proper score available from capture: the upper endpoint of a 2,000-draw whole-episode bootstrap interval for L1e minus frozen L1d cross-entropy to the exact recorded behavior distribution must be below zero. To prevent a tiny relative gain from promoting a poor model, validation $\mathrm{KL}(\mu\|\pi)$ must also be at most 0.1 nat. The diagnosed structure must be repaired directly: validation argmax agreement with the reactive action must be at least 0.95, and pooled agreement on focused- or lexical-decisive rows must be at least 0.5. All thresholds are frozen before fitting. The hard training target remains unchanged so this is one structural ablation; exact propensities are used for audit and evaluation only. L1e runs no Wine and can at most admit a separately preregistered integration canary.

L1e ran at solver commit `ec24e8ad05a6e25513ad0468e22bb53617ebc338`. It exhausted all 10,000 updates at final-to-initial train-gradient ratio 0.011684, above the frozen 0.01 threshold, so its immutable decision is `inconclusive-l1e-shared-action-optimization-not-converged`. Validation does not override that stopping decision. Reportable validation evidence was also uniformly unfavorable: $\mathrm{KL}(\mu\|\pi)$ was 0.986686 versus 0.563604 for L1d; the complete-episode L1e-minus-L1d exact-target cross-entropy interval was $[0.408315, 0.437710]$; reactive-action agreement was 0.655371 versus 0.748550; and focused/lexical final-tie agreement was exactly zero. Train KL 0.978077 and reactive agreement 0.660861 show the same pattern, so episode-split overfit is not a sufficient explanation. The plan, model, and result SHA-256 digests are respectively `c4c2ef915c158544f73ddc779acb2e28697c20e4e587081153d13421bc22b60d`, `8afc439fc6678a088fd0053086450d64f8344dc559382da116404759df5501f9`, and `a77b661136e3db85face5661c87919a13cf7966900b7b3fbd6a48cea6e2edce9`. No Wine canary ran.

The fitted normalized weights further localize the practical failure. The clearance and boundary-reserve weights were positive, 5.611145 and 8.651481, but stationary and focused weights were negative, -0.288433 and -0.633078, and lexical priority was nearly zero and negative. Those signs oppose the collector precisely inside its rare final tie strata. A global additive score cannot turn lower-priority fields off when a higher-priority field differs; their correlations over common non-tie rows compete with their conditional meaning. Thus sharing action weights alone does not implement lexicographic, non-compensatory semantics. Because the fit did not converge, L1e does not formally reject every shared linear fit; its wide performance deficit does reject this frozen run as a candidate and does not justify another post-hoc timebox extension.

More fundamentally, exact capture already supplies the behavior distribution, and portable current features already reconstruct its reactive winner on every row. Generic-network imitation of that known hand-coded controller is a model-class probe, not a necessary global prerequisite for learning factual HIT risk. The next bounded work therefore returns to the original diagnostic sequence: current-observation action-conditioned execution/dynamics probes, then observed-shield-collapse and fixed-horizon HIT probes. A fixed short history is admitted only if those expose temporal ambiguity. This reframing does not admit IQL or revise any L1–L1e decision.

The resulting L2 pilot is frozen in `experiments/l2-stage4-factual-probes-v1.json`. It reuses the immutable eight/four complete-episode split and runs no Wine. The alignment probe joins the first contiguous raw transition at each eligible decision root to its witnessed executed action and predicts factual $\Delta x, \Delta y$ using train-only per-action means. Its held-out executed-action joint MSE must be at most one quarter of the global-mean MSE. Publication/execution alignment must also be shown either by at least 99.9% direct agreement or, over at least 100 mismatches, by an executed-action MSE at most 80% of the published-action MSE. This disjunction distinguishes an exact publication path from a genuine pickup-delay population without post-hoc relabeling.

The two risk targets use fixed horizons of 1, 4, 16, and 64 physical game frames. A row exists only when every raw transition in its horizon is contiguous and free of observation-gap, Bomb, and infrastructure-failure exclusions. HIT is the disjunction of exact `life_lost` facts in that window. "Observed-shield collapse" is deliberately narrower than a future birth certificate: it is an explicit observed-shield control dead end or a strict contraction to a positive admissible-action count below the start count. A zero shield count on a passive or HIT root alone is not relabeled as collapse.

Both risk targets use the same 15 current-root/action features: player position and power; logarithmic bullet and laser counts; shield-set size; the published action's observed clearance knownness/value, projected endpoint and boundary reserve; equality to current input; and fixed movement direction and focus bits. A train-standardized ridge linear-probability probe is compared with the frozen train-prevalence constant by held-out Brier score. A horizon is eligible only with at least 20 train positives, 10 validation positives, and 100 validation negatives, and passes only when the upper endpoint of a 2,000-draw complete-episode bootstrap interval for candidate-minus-constant Brier is below zero. The pilot proceeds only if the alignment gate and at least one eligible horizon for each risk target pass. It fits no actor, defines no reward, and makes no gameplay claim. Regardless of outcome it does not itself admit history: a negative result first requires a separately frozen attribution among event scarcity, transparent current-root model misspecification, and temporal ambiguity; a positive result leaves history unsupported.

The pilot ran at solver commit `2c2af520b041804cf3a4036adc4d8fd0c3fe4175`. The train-only fit was serialized before the first validation load; the stored result has SHA-256 `4f1bad0d7f6720b4da55dda4927855b88f0249cf7f0ee95ff842255bb3b5ed07`, and an idempotent rerun reproduced that digest. The alignment view contained 193,331 train and 94,162 validation rows. Published action equaled the first witnessed executed action on every validation row. Executed-action $\Delta x, \Delta y$ joint MSE was 0.099123 versus 13.545947 for the global mean, a ratio of 0.007318; the decision-root current-action comparator remained at 12.486304. This both passes the frozen gate and rejects a one-frame action-label offset as an explanation of the earlier learner results.

The 1-frame HIT target had no positives and the 4-frame target had only 10 train and 3 validation positives, so both are insufficient rather than failed. At 16 frames the 542 validation positives yielded Brier 0.005573 versus 0.005737 for the train-prevalence constant, with a complete-episode candidate-minus-constant 95% interval of $[-0.0002393, -0.0001032]$. At 64 frames the corresponding 2,748 positives, Brier scores, and interval were 0.027221, 0.028629, and $[-0.0022318, -0.0008485]$. Thus both supported HIT horizons pass. All four shield-contraction horizons pass: their candidate-minus-constant interval upper endpoints are -0.012667, -0.040741, -0.092585, and -0.096139 at 1, 4, 16, and 64 frames respectively. The frozen decision is `proceed-current-observation-factual-signal`.

This result rejects only the claim that the portable current root lacks useful held-out factual risk signal for this transparent probe. Window labels overlap, so their positive counts are not unique HIT totals. Prediction under the logged action is not a counterfactual action-effect estimate, and the joint feature result does not yet isolate incremental action information from state information. The probe is not calibrated for deployment, is not an actor or critic, and has not run in Wine. The next bounded analysis should therefore decompose calibration, lifecycle/support, and state-only versus incremental action signal before freezing one confirmatory risk model; history and IQL remain unadmitted.

The first of those attributions is frozen in `experiments/l2b-incremental-action-v1.json`. L2b changes one surface: it removes the nine action-relative fields—clearance knownness/value, projected endpoint, boundary reserve, current-input equality, direction, and focus—while retaining the first six state fields, ridge penalty, labels, horizons, corpus, and split. It serializes the train-only state model before loading validation, reproduces the frozen full probe within 10^{-12} Brier, and asks whether the full-minus-state-only complete-episode Brier interval has an upper endpoint below zero at either already-supported 16- or 64-frame HIT horizon. Calibration bins, per-episode deltas, and shield-target comparisons are descriptive only. Because L2 already evaluated these four validation episodes, L2b is explicitly reused-heldout attribution rather than independent confirmation or promotion evidence. Either outcome still admits no history, value learner, policy, or Wine run.

L2b ran at solver commit `3e01cfd4c8cc0a255f9ff0209da32c21ae291d14`; its result SHA-256 is `e4a6e85ef9827a01a3f21aec97a137893defa09b8370b6c25180ca48ca7b1e83`. The full L2 Brier values reproduced with zero error. At 16 frames, full and state-only HIT Brier were 0.005572803 and 0.005601079, and the full-minus-state episode interval was $[-0.00005950, -0.00000047]$, passing just below zero. One of four per-episode deltas nevertheless favored state-only. At 64 frames, full and state-only Brier were 0.027221173 and 0.027271775, but the interval $[-0.00048632, 0.00031167]$ crossed zero and two episodes favored state-only. The full model's 10-bin HIT calibration error was also worse: 0.00204 versus 0.00096 at 16 frames and 0.00520 versus 0.00111 at 64 frames.

Action-relative geometry contributed much more consistently to the auxiliary shield target. Full-minus-state-only shield-contraction Brier intervals were strictly negative at all four horizons, and every individual validation episode favored the full representation. The formal L2b decision is `proceed-action-relative-current-root-signal` because its frozen rule required one supported HIT horizon. The substantive claim is deliberately narrower: the result rejects "all signal is state-only," but action-relative physical-HIT information is small, horizon-dependent, heterogeneous, and not yet calibrated. Independent confirmation plus support/lifecycle decomposition must precede any value learner or online candidate.

That decomposition is frozen in `experiments/l2c-stage4-boundary-diagnosis-v1.json`. L2c fits nothing and reuses the same four validation episodes only to retain recorder facts beside the exact L2 rows: the published action's logged propensity, equality to the reactive baseline, shield-set size, and elapsed game frames since the latest physical HIT. These are diagnostic strata, not actor inputs. In particular, an action unequal to baseline proves a nonbaseline draw, whereas equality does not identify the latent 80/20 mixture component because uniform exploration can sample the baseline action. HIT never terminates or removes the remainder of an episode. Ten fixed equal-width reliability bins, calibration in the large, ECE, and clipping counts diagnose probability scale but do not calibrate or select a model.

L2c ran at solver commit `c5f03db73ba8987d88d54ffff19931779b2fa578`; its result SHA-256 is `bf2fa3d74b16f13102a25a7f809e56bfb0841b2955f2249b9c62b399dd3276cc`. It reproduced every 16/64-frame L2b full and state-only Brier value with zero error. At 16 frames, the overall full-minus-state-only delta remained -2.83×10^{-5}, but the 76,216 baseline-equal rows had delta -3.63×10^{-5} while the 17,706 rows known to differ from baseline had $+6.38 \times 10^{-6}$; only two of four episodes favored full in the latter stratum. At 64 frames the corresponding deltas were -7.43×10^{-5} and $+5.17 \times 10^{-5}$, with only one of four nonbaseline episode views favoring full. Thus the prior increment is concentrated in baseline-equal behavior, consistent with reactive tie-breaking or route context rather than a supported physical-HIT action value.

The 16-frame pre-first-HIT view still had a negative -2.16×10^{-5} delta and all four episodes favored full, so a purely post-HIT lifecycle artifact is not supported. Calibration remains a separate failure: at 16 frames the full versus state-only calibration-in-the-large and ECE were 0.00204 versus 0.00096, and 36,406 versus 34,013 of 93,922 predictions clipped to zero. The formal L2c decision is `freeze-fresh-independent-confirmation`. That confirmation must use untouched models and wholly new complete episodes, and report overall replication, nonbaseline support, pre-first-HIT lifecycle, and calibration as separate gates. L2c identifies no causal effect and admits neither history nor value learning.

The independent confirmation is frozen in `experiments/l2d-stage4-fresh-confirmation-v1.json`. It collected eight new serial natural-RNG Stage 4 episodes under eight new policy RNG seeds and the unchanged reactive plus 20% uniform observed-shield mixture. Collection could not peek or stop sequentially, physical HIT could not cause a retry, and the untouched L2/L2b fits were evaluated only after all eight slots completed. The primary gate required 16-frame event support, at least six of eight episode directions favoring full, and a 2,000-draw complete-episode bootstrap upper endpoint below zero. Separate point gates required known-nonbaseline and pre-first-HIT rows from every episode with frozen positive-count minima. Calibration readiness required absolute calibration-in-the-large at most 0.002 and full ECE no more than 0.001 above state-only. Passing predictive gates could not admit causal action value, a learned policy, or gameplay claims.

L2d ran at solver commit `c17839cbaed0643e567193ee8dc2e630931b70cf`. All eight episodes were accepted on attempt one, with 91 physical HITs total, zero Bomb, and no rejected infrastructure attempt. The collection ledger and result SHA-256 values are `5c9625b5ea82e2ef033b663e9a68535df13e6e7793673d6c326b0b7463bfb50c` and `ee240dc3a8bfa5a02eaa146f7a37ef9cbf281c39edf7712b7f79046911e0ad2f`. The primary view contained 188,791 rows and 1,085 positive windows. Full versus state-only Brier was 0.0055114 versus 0.0055610; all eight episode directions favored full, and the full-minus-state interval was $[-7.14 \times 10^{-5}, -2.75 \times 10^{-5}]$. Thus independent signal replication passed.

The known-nonbaseline view contained 35,272 rows and 172 positives from all eight episodes. Its point delta changed from the old positive value to a weak -2.10×10^{-6} and therefore passed the frozen point/support boundary, but only three of eight episode directions favored full. In contrast, the baseline-equal delta was -6.05×10^{-5} and favored full in every episode. The pre-first-HIT delta was -2.00×10^{-5} with all eight directions favorable, independently rejecting a purely post-HIT explanation. At 64 frames the descriptive overall and nonbaseline deltas were -4.56×10^{-4} and -2.39×10^{-4}, with seven and six favorable episode directions respectively.

Calibration readiness failed. At 16 frames, full calibration-in-the-large was 0.00193 and met its ceiling, but full ECE 0.001976 exceeded state-only ECE 0.000960 by 0.001016, just above the frozen allowance; 73,334 full predictions clipped to zero. At 64 frames the ECE gap was much larger, 0.005139 versus 0.000253. The formal decision is `confirm-predictive-signal-calibration-not-ready`. Fresh data therefore confirm incremental logged-action predictive information, not a stable nonbaseline ranking, causal action effect, calibrated action value, or online candidate. The next bounded experiment remains offline and may change only the train-only probability/calibration surface.

That experiment is frozen in `experiments/l2e-train-only-calibration-v1.json`. L2e changes only the probability surface: a two-parameter affine-logistic Platt mapping is fit on the original eight L2 train episodes over the frozen, unclipped h16 full ridge score. The mapping has no actor inputs of its own and cannot change the ordering of that scalar score. The fit was serialized before the already inspected L2d episodes were loaded. Those eight episodes provide a one-time selection evaluation, not independent confirmation. The primary gates require complete-episode Brier improvement over both uncalibrated full and state-only, preservation of the known-nonbaseline and pre-first-HIT point boundaries, and the unchanged L2d calibration limits.

L2e ran at solver commit `9004e387249f1a1b3a2c2c92e3da05333a470733`. The fit and result SHA-256 values are `ae3e25a2b440b5afc8aa83dacf5f71f8c260c88898694ee0bd651d7c405e7720` and `dbe9d9bb58f442ec14c3e0ea2e2416ab68e4e3d7affec2c052c9ae7fadb0a9d8`. The damped-Newton fit converged in five accepted steps with final gradient infinity norm 5.73×10^{-11}; its raw-score slope was positive. Source reproduction was exact.

On the 188,791 reused L2d rows, calibrated-full Brier was 0.0056111, worse than uncalibrated full at 0.0055114 and state-only at 0.0055610. The calibrated-minus-uncalibrated interval was $[-2.62 \times 10^{-5}, 2.21 \times 10^{-4}]$, and the calibrated-minus-state interval was $[-8.07 \times 10^{-5}, 1.78 \times 10^{-4}]$; both point estimates were positive. Known-nonbaseline and pre-first-HIT calibrated-minus-state point deltas were likewise positive, and only two and three of eight episode directions favored calibrated full. In contrast, calibration-in-the-large improved from 0.001931 to -0.000185 and ECE improved from 0.001976 to 0.001553, so the unchanged calibration-readiness limits passed.

The formal decision is `reject-train-only-platt-calibration`. The surface slightly improved reused-evaluation logistic NLL while regressing the preregistered Brier target. It removed clipped zeros but expanded the number of predictions at least 0.1 from 26 to 1,404; most populated higher bins overpredicted factual event rates. Thus a global logistic remapping of this linear-probability score is not sufficient. No post-hoc calibration family, fresh confirmation, history, value learner, or online policy is admitted. A future direct train-only probability model would be a new experiment and must align its training proper score with its selection proper score.

That direct model is preregistered in `experiments/l2f-action-conditioned-hazard-v1.json`. L2f is not behavior cloning and does not transform a frozen ridge score. It fits one shared gradient-boosted regressor to the factual indicator of physical HIT within 16 game frames under logged behavior continuation. Each row contains the same 15 portable current-root/action facts used by L2; the only target action is the one Wine actually executed. The comparison fit removes exactly the nine action-relative fields while retaining the identical model family. Neither fit uses propensity, baseline identity, lifecycle, source facts, future facts, or a counterfactual successor as input.

The capacity and optimizer are fixed before fitting: 64 depth-3 CPU histogram trees, learning rate 0.05, minimum child weight 64, unit row/column sampling, single-thread execution, fixed seed, and no validation early stopping or parameter sweep. Squared-error training is the unweighted row-level Brier score used for selection. Predictions are bounded to $[0, 1]$, with a frozen 0.001 maximum raw clipping fraction so clipping cannot silently become the probability model. This tabular conditional model is the smallest active architecture that can express state–action interactions already suggested by the factual geometry; it is not claimed to be a long-horizon value function.

The model is fit on the original eight L2 train episodes and serialized before the eight already inspected L2d episodes are loaded. Those L2d episodes can only select a fresh confirmation. Selection jointly requires complete-episode Brier improvement over frozen L2 full and over the same-architecture state-only fit, at least six of eight favorable overall episode directions, a bootstrapped known-nonbaseline increment with at least five favorable directions, a bootstrapped pre-first-HIT increment with at least six, sufficient frozen event support, and the unchanged L2d calibration limits. Action and propensity strata are reported as diagnostics. Passing does not establish causal action value or admit a policy, history, value learning, or Wine; it freezes this scorer for wholly new complete-episode confirmation.

L2f was executed once at commit `7289949466c4ce38b3f1b43d853fa4bd72e3d02d`. The plan, train-only fit, and final result SHA-256 hashes were respectively `13060ab55c2095f96587d894efba57f11d599de3120` `fdc6f47843fc55defaf0989d2aac9dd613c1ce270c29fe48d1dd87a1dc85eb64`, and `58667fa7e535c30522aedde1f2f` The fit artifact recorded that evaluation had not been loaded before either model was serialized. Frozen L2 source scores were reproduced exactly.

On 188,791 reused-L2d rows with 1,085 positive windows, direct full Brier was 0.00480785, versus 0.00551139 for frozen L2 full and 0.00500959 for the direct state-only ablation. Direct full minus frozen full had a complete-episode bootstrap interval of $[-9.13 \times 10^{-4}, -4.98 \times 10^{-4}]$; direct full minus direct state-only had $[-2.54 \times 10^{-4}, -1.52 \times 10^{-4}]$. Average precision rose to 0.2816 from frozen L2's 0.0772, ROC AUC rose to 0.9661 from 0.9276, and both calibration-in-the-large at -0.000248 and ECE at 0.001368 passed. Thus direct nonlinear Brier fitting, overall incremental action signal, support, and calibration all passed. This is positive evidence for the model family, not for an online policy.

The immutable decision was nevertheless `reject-action-conditioned-h16-hazard`. The full candidate itself clipped only 0.0773% of raw evaluation predictions, below the 0.1% limit, but the frozen joint bounded-surface gate also required the state-only comparator, which clipped 16.70%. On 35,272 known-nonbaseline rows, full minus state-only was favorable at -2.19×10^{-5} in six of eight episodes, but its interval narrowly crossed zero at $[-4.79 \times 10^{-5}, 4.66 \times 10^{-6}]$. On 34,154 pre-first-HIT rows the point delta was -3.59×10^{-5}, but only four of eight episode directions favored full and the interval also crossed zero. No fresh confirmation, Wine canary, history model, or value learner was run.

The action names themselves were supported: all 18 occurred with positive windows in all eight episodes. The sharper boundary was their logged probability. Below propensity 0.025, 34,792 rows and 158 positives had a full-minus-state point delta of $+1.12 \times 10^{-6}$; at propensity at least 0.5, 153,519 rows and 913 positives had -2.43×10^{-4} with all eight episode directions favorable. The unweighted objective is therefore dominated by the reactive baseline measure, whereas deployment would rank the entire observed-shield set. This selected action-measure mismatch as the next falsifiable explanation; it was not yet evidence that current-root observations require history, and L2g subsequently rejected it as the primary cause.

The research decision is to modify the objective measure while retaining the architecture. L2g is preregistered in `experiments/l2g-uniform-shield-weighted-hazard-v1.json` and uses the exact one-step ratio $w = (1/|\mathcal{A}_{\text{shield}}|)/\mu(a \mid s)$ in both fitting and its primary Brier comparisons. Under the frozen 80% baseline plus 20% uniform collector, a nonbaseline action has weight exactly five, so this is bounded action-distribution correction rather than long-horizon OPE. Propensity and baseline identity remain offline-only. Shield-set size retains its existing portable current-root feature role and also defines the target probability, so the feature schema is unchanged. All weights must be finite, positive, at most five, and reproduce the declared collector formula; the full train and reused-L2d inventories remain unchanged.

A pre-fit read-only contract audit reconstructed all weights without fitting a model. The 192,791 train rows had range $[0.068493, 5]$, mean 1.00113, and effective sample size 41,048; the 188,791 reused-L2d rows had the same range, mean 0.99486, and effective sample size 39,960. Maximum absolute error against the declared collector probability was 6.94×10^{-18} and every known- nonbaseline weight equaled five within 8.88×10^{-16}. This validates the bounded one-step weighting contract but also records its roughly 40,000-row effective support; it is not a model result or independent evidence.

L2g retains L2f's 64 depth-3 trees, learning rate, regularization, h16 factual target, portable 15-field current-root/action vector, and state-only ablation. It serializes both weighted fits before loading reused L2d. Its primary uniform-target episode bootstrap must put the upper endpoint below zero for weighted full versus both frozen unweighted L2f full and weighted state-only, with at least six of eight state-ablation episode directions favorable. The known-nonbaseline and below-0.025 propensity views must each have sufficient factual positives, all eight episodes, an upper endpoint below zero, and at least six favorable episode directions. The unchanged pre-first-HIT support must also have an upper endpoint below zero and at least six favorable directions. Target-weighted calibration-in-the-large must be within 0.002 and target-weighted ECE may exceed weighted state-only by at most 0.001; logged- measure calibration-in-the-large retains the same 0.002 guardrail.

L2f showed that applying a clipping gate to a noncandidate comparator can reject an otherwise bounded full candidate. L2g prospectively corrects that gate: only the selected full model must keep raw clipping at or below 0.001; comparator clipping remains fully reported. Passing selects the immutable scorer for wholly fresh confirmation only. If this corrected current-root experiment still fails the low-propensity/nonbaseline and pre-first-HIT gates, one fixed short history becomes the next representation test. Tree sweeps, scalar calibration, collector BC, IQL, and online evaluation are not responses to the L2f result.

L2g was executed once at commit `adb355875a0c008fc057881bad2c7f99e08e4302`. The preregistration, plan, train-only fit, and final result SHA-256 hashes were respectively `e893a608276d26d8dbc75dbf73da037` `64d4a8e46844a590fb8c8f3524ce9e1eec64d21219a934a0270b955d6eeadb52`, `e49143359a17e440221964a9977cc81` and `81dc532d094199133f72918f1c23931c614ae478c05770197c921ce4c523b9c7`. The weighted fits were serialized at 10:52:43 UTC before the reused-L2d load. Frozen L2f Brier and NLL reproduced with absolute error zero.

On train, target-weighted full Brier was 0.00447703, versus 0.00459510 for weighted state-only and 0.00541167 for constant prevalence. The intervention therefore learned a train action increment. Its unconstrained surface was already defective: 13.38% of full predictions and 12.23% of state-only predictions required clipping below zero before evaluation data loaded.

On reused L2d, target-weighted full Brier was 0.00454350, versus 0.00455238 for weighted state-only and 0.00449955 for frozen unweighted L2f full. Weighted full minus frozen L2f full had point $+4.40 \times 10^{-5}$, interval $[-3.48 \times 10^{-5}, 1.28 \times 10^{-4}]$, and four of eight favorable episode directions. Weighted full minus weighted state-only had point -8.87×10^{-6}, interval $[-5.54 \times 10^{-5}, 3.64 \times 10^{-5}]$, and four favorable directions. Under the logged measure, weighted full Brier worsened from L2f's 0.00480785 to 0.00493361 with a wholly positive interval. Thus the exact weighting change failed both its primary target-measure comparator and the overall action increment.

The bulk below propensity 0.025 had 34,792 rows, 158 positives, a full-minus-state point of $+2.11 \times 10^{-5}$, interval $[-2.56 \times 10^{-5}, 6.39 \times 10^{-5}]$, and only three favorable episodes. Known-nonbaseline had 35,272 rows, 172 positives, point $+3.96 \times 10^{-6}$, and four favorable episodes. Pre-first-HIT retained a favorable point of -4.78×10^{-5} but its interval crossed zero and only five episodes favored full. In contrast, baseline-equal rows retained a -2.06×10^{-4} point increment with seven favorable directions. Exact action-measure correction therefore did not remove the baseline-equal pattern.

The full evaluation surface clipped 13.53% of raw predictions and failed its candidate-bound gate. Target-weighted calibration-in-the-large at 1.76×10^{-5}, target-weighted ECE at 0.000823, and logged calibration- in-the-large at -4.87×10^{-5} all passed. Aggregate calibration after clipping does not repair the invalid raw surface or failed proper-score gates. The immutable decision is `reject-weighted-h16-hazard`; no fresh confirmation, Wine canary, history fit, or value learner ran.

At the research level, importance weighting as the sole repair is stopped. Action-measure mismatch may contribute variance, but is not the primary L2f root cause because the exact correction failed its own target score and exploration-action views. A bounded link could remove clipping but cannot by itself explain those failures. Following the frozen L2g boundary and the TH06 update-order partial observability result, the next separately preregistered experiment adds one fixed short portable history under the unweighted L2f measure. It must preserve causal root order and the same low-propensity, nonbaseline, and pre-first-HIT views. If history fails, the next boundary is the information discarded by the bounded current-root object summary, not a weight sweep or immediate value learner.

L2h is preregistered in `experiments/l2h-fixed-history-hazard-v1.json`. A history-ready row has exactly 16 immediately preceding, consecutive, learning-eligible policy roots. Every one of those prior root intervals must span exactly one game frame; a non-unit interval resets the history rather than bridging a HIT, respawn, or capture gap. Each past root contributes the unchanged 15 portable L2 fields evaluated for the action actually published at that past root; fields are ordered oldest to newest before the current 15-field action-conditioned root. A row without the full prefix is excluded rather than padded. No prior outcome, HIT/lifecycle indicator, behavior propensity, baseline identity, absolute frame, source fact, or future fact is present. At online inference the same ring buffer can be updated only after a published action, so the representation has an exact causal export contract.

All three L2h fits retain L2f's 64 depth-3 trees, learning rate, regularization, deterministic 16-thread CPU execution, unweighted h16 Brier objective, and train/evaluation episode identities. The history-full fit uses all 255 fields. The current-only comparator uses the final 15 fields on exactly the same history-ready rows. The current-action-ablated-history comparator retains all 240 past fields plus the six current state fields and removes exactly the nine current action-relative fields. Thus the first comparison tests temporal information and the second tests current-action discrimination conditional on that history, without changing model family or row support.

The train-only artifact is serialized before reused L2d loads. Selection requires complete-episode Brier intervals below zero for history-full minus current-only and history-full minus current-action-ablated history, each with at least six favorable episode directions. The latter action comparison must also pass below-propensity-0.025, known-nonbaseline, and pre-first-HIT support views with intervals below zero and at least six favorable directions. The full candidate alone must satisfy the 0.001 raw-clipping limit; calibration-in- the-large must be within 0.002 and its ECE may exceed the action-ablated comparator by at most 0.001. Passing selects wholly fresh complete-episode confirmation only. If temporal gain and exploratory action signal remain absent, a bounded current-root object-set representation becomes the next candidate; history length, tree, weighting, and value-method sweeps remain unadmitted.

A read-only pre-fit audit was completed before freezing or evaluating any L2h model. The eight train episodes contain 192,791 factual h16 current rows; 190,469 retain a valid history, with 1,105 positives. The eight reused L2d episodes contain 188,791 current rows; 186,615 retain a valid history, with 1,075 positives. Hence the reset rule drops only 2,322 train rows and 2,176 evaluation rows. On evaluation history-ready rows there remain 156 positives below behavior propensity 0.025, 170 known-nonbaseline positives, and 96 pre-first-HIT positives, all above their frozen support thresholds. The history elapsed-game-frame minimum, median, and maximum are all exactly 16 in both splits. The all-current frozen-L2f evaluation Brier also reproduces at 0.004807849266974402. A repeated 16-thread fixture fit serialized identically; thread count is frozen for runtime only and is not an algorithmic ablation.

L2h ran once from commit `bbf6e81a3203fb00443060a215949a37fdb13352`. The immutable result SHA-256 is `b91e44250b3c5a8551a861d978b1d8558b050422f5e596a3280568d35a2af37b`; the frozen L2f Brier and NLL reproduced with zero error. History-full evaluation Brier was 0.00479407, versus 0.00483026 for same-row current-only, 0.00488334 for history with current action fields ablated, and 0.00482784 for frozen L2f on the same rows. Its full-minus-current-only point delta was -3.62×10^{-5}, but the complete-episode 95% interval $[-1.44 \times 10^{-4}, 1.02 \times 10^{-4}]$ crossed zero and only five of eight episodes favored history. The temporal-gain gate failed.

Current action fields did add an aggregate increment conditional on history: full minus action-ablated Brier was -8.93×10^{-5} with interval $[-1.05 \times 10^{-4}, -7.03 \times 10^{-5}]$ and all eight favorable episode directions. This increment was again baseline-heavy. All eight baseline- equal episode directions favored full, with point delta -1.08×10^{-4}. Below propensity 0.025 the point was -5.18×10^{-7} with interval $[-4.10 \times 10^{-5}, 3.97 \times 10^{-5}]$ and three favorable episodes; known-nonbaseline was -7.95×10^{-6} with $[-4.51 \times 10^{-5}, 2.74 \times 10^{-5}]$ and four; pre-first-HIT was -2.49×10^{-5} with $[-5.36 \times 10^{-5}, 1.40 \times 10^{-6}]$ and five. All three frozen boundary gates failed despite sufficient support.

The history-full raw surface clipped 4,543 of 186,615 rows (2.434%), failing the 0.001 candidate limit. Calibration-in-the-large was -0.000135 and ECE was 0.001735, so aggregate calibration passed and does not explain away the unbounded surface or heterogeneous boundaries. Post-hoc feature-gain accounting assigns 37.78% to prior roots, ruling out a silently unused history, but most history gain is sparse player-position, power, and shield-count context rather than retained hazard geometry. The formal decision is `reject-fixed-history-h16-hazard`; no fresh confirmation, Wine run, policy, or value fit occurred.

L2h therefore rejects this fixed scalar history, not all temporal information. A bounded link could repair range violations but cannot make the temporal or exploration-action intervals negative. The next high-information boundary is a separately preregistered bounded current-root set encoder over captured instantiated bullets, lasers, and enemy bodies. It must freeze its cap, order, mask, portable action-relative geometry, scalar-only and current-action- ablated comparators, and immutable inference-cost bound before fitting. It may not use future births, source identity, an invented successor, value learning, or Wine evaluation.

L2i asks whether the information missing from the scalar root is the individual geometry of already-instantiated hazards. A read-only audit preceded the preregistration. Exhaustive occupied-bullet counts have median 193/195, 99th percentile 630/631, and maximum 640 on train/reused-L2d rows, so flattening the raw pool is rejected. In contrast, the exact four-frame primitives already recorded for the observed shield yield spatiotemporal-token median 1, 90th percentile 6, 95th percentile 16, 99th percentile 27, and maxima 41/42. All 192,791 train and 188,791 reused-L2d h16 rows have the expected contract, four-frame shape, finite geometry, and zero malformed tokens. A cap of 48 therefore causes zero truncation in both audited splits. Positive rows contain 5.307/4.994 tokens on average versus 2.564/2.577 for negatives; this is support for testing the representation, not a learner result or reward.

The L2i token is deliberately source-free and correspondence-free. Each AABB or laser primitive at each of the four shield frames becomes one token with a two-way type indicator, four-way frame indicator, anchor or center relative to the current player, and normalized physical geometry. AABB geometry contains half width and height; laser geometry contains cosine and sine of angle, its two recorded extent values, and width. Fixed divisors derived from the portable playfield/geometry contract are used instead of train-fitted normalization. No bullet state, EX flag, sprite, slot, enemy program, ECL, RNG, address, source identity, future birth, or object correspondence is read. The set is action-independent; the factual published action conditions the encoder through the same current portable action fields used by L2f. If a future root exceeds 48 tokens, one frozen frame/type/geometry ordering truncates it and the event is counted; no audited train or evaluation row exercises that fallback.

L2i is preregistered as `experiments/l2i-observed-primitive-set-hazard-v1.json`. It fits three identical cap-48 sigmoid DeepSets from one shared initialization and identical minibatch order: object-full, scalar-only with the set mask removed, and object set with exactly the nine current action-relative scalar fields removed. The shared token MLP has two width-32 ReLU layers; masked mean and maximum pooling, normalized log token count, and the 15 scalar slots feed width-64 and width-32 ReLU head layers. The ablations retain the same tensor shapes and parameter count by zero masking. Training uses float32, direct unweighted row Brier, AdamW at 0.001 with weight decay 0.0001, batch size 4096, 40 fixed epochs, seed 20260846, eight order-preserving episode-loader workers, 32 deterministic CPU fit threads, and no validation early stopping, weighting, calibration fit, or model sweep. A train-only artifact is frozen before the already inspected L2d episodes load.

Selection requires zero primitive-contract failures and truncations, a complete-episode Brier interval below zero for object-full minus scalar-only, and another below zero for object-full minus current-action-ablated object set, each with at least six favorable episode directions. The current-action comparison must separately pass the unchanged below-propensity-0.025, known-nonbaseline, and pre-first-HIT support/interval/direction gates. Sigmoid probabilities must be finite with saturation fraction at most 0.001; calibration-in-the-large must be within 0.002 and full ECE may exceed the action-ablated comparator by at most 0.001. The immutable scorer must remain below 10,000 parameters and batch-18 one-thread CPU p99 inference below 5 ms. Frozen L2f all-row scores must reproduce exactly. Passing selects a wholly fresh confirmation experiment only; it does not establish causal action value, authorize exhaustive objects or history, fit a policy/value, or run Wine.

L2i ran once from commit `b8344f60e3d0d654e1bc23b2108cf4ea7231cd9a`. The immutable decision is `reject-observed-primitive-set-h16-hazard`. The 188,791 evaluation rows contained 1,085 positives, maximum 42 tokens, zero truncation, and exact frozen L2f Brier/NLL reproduction. The 8,929-parameter scorer met its export bound with batch-18 one-thread p99 0.388 ms. Object-full Brier was 0.00473906 versus 0.00468783 for scalar-only. Their difference was $+5.1231 \times 10^{-5}$, episode-bootstrap 95% interval $[-9.6905 \times 10^{-5}, +2.3328 \times 10^{-4}]$, with five of eight favorable directions; object information therefore failed its gate. Object-full versus frozen L2f was -6.8787×10^{-5}, but its interval also crossed zero.

The aggregate current-action comparison did pass: full minus action-ablated Brier was -1.2621×10^{-4}, interval $[-1.9912 \times 10^{-4}, -6.1912 \times 10^{-5}]$, with all eight episodes favorable. It again localized to baseline-equal behavior. Below propensity 0.025 the delta reversed to $+2.2964 \times 10^{-4}$ with a wholly positive interval and two favorable episodes; known-nonbaseline was $+1.6414 \times 10^{-4}$ with an interval crossing zero and two favorable episodes; pre-first-HIT was $+2.2714 \times 10^{-4}$ with a wholly positive interval and two favorable episodes. These are not supported action-value effects.

The fitted probability surface supplies a separate algorithmic diagnosis. Object-full assigned 122,814 rows (65.05%) probability at most 10^{-7} and reached logit -38.09. Its mean prediction was 0.002653 against event rate 0.005747, so calibration-in-the-large was -0.003094 and failed the 0.002 bound. Yet its average precision 0.31373 exceeded scalar-only 0.29665, consistent with geometric ranking information but a bad probability surface. For sigmoid probability p and logit z, direct Brier has derivative $2(p-y)p(1-p)$; its correction on a rare missed positive vanishes as p tends to zero. Unweighted Bernoulli log score instead has derivative $p-y$ and the same population probability target. The next separately frozen experiment therefore changes only L2i training loss to BCE-with-logits. It retains every row, token, architecture, optimizer, epoch, comparator, and evaluation gate; it is not scalar calibration, class weighting, a loss sweep, or admission of object sets, value learning, fresh confirmation, or Wine.

L2j is preregistered in `experiments/l2j-logscore-primitive-hazard-v1.json`. Its unweighted `BCEWithLogitsLoss` uses positive and negative class weights exactly one. The L2i initialization, minibatch permutations, AdamW settings, 40 epochs, batch size 4096, seed 20260846, 8,929-parameter architecture, cap-48 tokens, eight loader workers, and 32 fit threads remain exact. All L2i object-information, action, support, lifecycle, calibration, saturation, and export gates remain. One additional complete-episode bootstrap compares L2j object-full against frozen L2i object-full using seed 20260853; its upper 95% endpoint must be below zero with at least six of eight episode directions favorable. Frozen L2f and L2i Brier/NLL must reproduce within 10^{-12}. The train artifact is serialized before reused L2d loads. Passing every gate can select wholly fresh confirmation only; no validation checkpoint selection, new Wine data, policy, value model, or online run is admitted.

L2j ran once from commit `90d3988f565c9d3972b433a392f4f0b7ca17e11a`. The immutable decision is `reject-logscore-observed-primitive-h16-hazard`. Frozen L2f and L2i Brier/NLL reproduced with zero absolute error. The preregistration, frozen plan, train-only fit, and result SHA-256 digests were respectively `d14db15d9bdc316c`, `3cd5544fda8ca808`, `fcc1e63a6f8721af`, and `2d6006178d01f526`; the full digests are retained in the per-experiment research log. Object-full Brier was 0.00464423, NLL 0.01973721, average precision 0.33083, and AUC 0.97269; scalar-only Brier was 0.00463179. Full minus frozen L2i Brier was -9.4838×10^{-5}, complete-episode interval $[-2.0763 \times 10^{-4}, -9.3637 \times 10^{-6}]$, but only five of eight episode directions were favorable. Full minus scalar-only was $+1.2431 \times 10^{-5}$ with an interval crossing zero and four favorable episodes. Overall full minus action-ablated was -3.5818×10^{-4}, interval $[-4.6076 \times 10^{-4}, -2.5569 \times 10^{-4}]$, with all eight favorable. Low-propensity, known-nonbaseline, and pre-first-HIT intervals crossed zero with only two, four, and five favorable episodes respectively; baseline-equal again passed in all eight.

BCE reduced full probabilities at most 10^{-7} from 65.05% to 37.07%, NLL from 0.02866 to 0.01974, and calibration-in-the-large magnitude from 0.003094 to 0.002401. The frozen row-fraction saturation and 0.002 calibration limits still failed. A read-only outcome audit gives the mechanism a sharper interpretation: L2i's saturated rows contained 89 positives, whereas L2j's contained one. BCE repaired almost all confident missed positives, so its rare-positive gradient hypothesis was real but insufficient. This post-hoc fact does not revise the formal rejection; future probability gates should distinguish saturated easy negatives from saturated positives.

The more fundamental failure was target attribution. On all 188,791 reused evaluation rows, the initial action had median run length one frame, changed on the second frame in 74.46%, and the median first-HIT offset was 11. Of 1,085 h16 positives, 1,079 (99.45%) first hit after that action run ended. For low-propensity rows the action changed on frame two in 91.97% and 157/158 positives hit afterward; known-nonbaseline was 91.71% and 170/172. Only six overall positive rows hit during their initial action. Thus aggregate action ablation primarily identifies correlation between collector mode/action and a later factual behavior continuation. It cannot support a claim about the effect or value of the usually one-frame current action.

Further loss, width, pooling, attention, history, calibration, or weighting sweeps on this action-conditioned h16 target are stopped. The next separately preregistered feasibility experiment must create measurable factual exposure: hold a randomized observed-shield-admissible action intention for one fixed short duration, recheck the observed shield at every coherent root, and record all conditional propensities and overrides. It must preserve zero Bomb, lifecycle, exact PID/input delivery, root linkage, normal clock, and complete episodes. Passing that collection contract can define an exposure-conditional physical-HIT target; it does not invent a counterfactual successor, widen the shield, admit an object encoder or value learner, or authorize a Wine policy canary.

L2k freezes this boundary in `experiments/l2k-stage4-action-exposure-v1.json`, SHA-256 `9e2b2adc540b7e9d8493850ec43948d62c193b66d8f7e57e14403affbdaba94b`. A read-only sizing audit over the eight immutable L2d episodes measured the fraction of a uniformly selected initial action that remained in every later factual shield set. Action-weighted persistence was 97.57%, 93.62%, and 87.43% for two, four, and eight policy roots. These are behavior-continuation mask facts, not counterfactual successors or safety evidence. Four roots is the smallest tested exposure that materially changes the old median-one-frame action while retaining high observed-mask persistence; duration is not swept online.

The boundary ownership is explicit. The policy owns a run-local random group assignment and its four-step intention. The controller remains authoritative for a freshly recomputed observed shield, exact publication, input pickup, and lifecycle interruption at every root. The recorder stores group ID, step, intended action, initial assignment probability and full distribution, and any shield override independently of learner features. The episode loader checks raw/transition agreement and reads both immutable v13 episodes and new v14 exposure episodes. These identifiers and propensities are provenance, not actor inputs.

At step zero the assignment distribution is exactly uniform over the current observed-shield set. At steps one through three, the conditional published distribution is a point mass on the intention if it remains admissible, or on the current reactive baseline with an explicit override reason otherwise. The shield is still replayed after policy work. A target already inside the bounded input-pickup lease that loses admission is released and remains the existing fail-closed control-dead-end; L2k does not manufacture a successful override after partial delivery. Physical HIT, passive control, inactive player state, and control dead-end interrupt an unfinished assignment without entering the actor observation.

The first experiment fixes two serial normal-speed natural-RNG complete Lunatic Stage 4 episodes, both pilot-train-only. Each must contain at least 4,000 eligible complete groups; all 18 actions need at least 100 aggregate assignments; at least 80% of eligible complete groups must have no override and at least four witnessed intended-action executions; control dead-end rate must be at most 0.5%; and metadata, Bomb, infrastructure, linkage, and complete-episode gates must pass exactly. H16-positive group starts have a 64-example readiness diagnostic, but it is deliberately not a retry or contract gate: good play or natural low HIT support cannot be converted into selection by recollection. Passing with support selects only a separate stateful serial-versus-parallel differential and scaled train collection. No fit, learned online policy, fresh evaluation, or policy-improvement claim is admitted.

Both fixed pilots completed on their first attempt. Episode `20260822T135910Z-643218400` recorded 26,855 transitions, 18 physical HITs, and zero Bomb; episode `20260822T141219Z-21576 0100` recorded 26,976 transitions, 12 physical HITs, and zero Bomb. Both passed complete dataset admission, dense native shield parity, bit-exact player-successor replay, zero policy callback failures, and their frozen latency limits. The aggregate exposure audit measured 12,936 eligible complete groups, 645–815 assignments for every action, no-override and four-intended-execution fractions of 0.93677, a control-dead-end rate of 0.00217, and 92 h16-positive group starts. All numeric, Bomb, and infrastructure gates passed.

The immutable as-run result nevertheless recorded `reject-action-exposure-collection-contract`. Its exact-contract checker classified five partial groups as unexplained interruptions. A read-only transition-level diagnosis found that every case contained a factual `outcome.control_dead_end` transition followed by a new step-zero group. Control dead-end was already one of the frozen lifecycle interruptions; the checker had accepted only HIT, non-learning-eligible, or terminal collapsed epochs and therefore missed the case where an earlier published action was witnessed before the dead end. This is an audit implementation defect, not a collector exception.

The original result is retained unchanged at SHA-256 `f40ec816faff8340ba029f500c6de23923 9215695e02113909bda481733891fb`. A bounded erratum adds only the factual control-dead-end transition condition, changes no episode, metric, threshold, assignment, or label, and reproduces all other audit fields exactly. Its separate artifact, `artifacts/l2k-stage4-action-exposure-v1/audit-erratum-v1.json`, has SHA-256 `97c51277f35c234c2673b750ad040b359760b82f0b30a2cae4fcf89a3124b446` and the corrected decision `proceed-action-exposure-training-collection`. This admits the two episodes only as pilot-train data and selects a separately frozen stateful serial-versus-parallel differential. It does not admit scaled parallel collection, hazard fitting, value learning, a Wine canary, or a gameplay claim.

L2k passing the recorder contract did not by itself make its target useful. A strict four-unit-frame view requiring witnessed intended-action execution had zero HIT positives among 12,985 group starts. Treating the randomized intention as the intervention and retaining the controller's deterministic dead-end/input-release response admitted only four positives among 12,948 examples; all four followed a control dead-end. Scaling the same protocol would therefore produce an almost entirely negative action target.

A train-only, read-only duration audit measured nonoverlapping factual behavior-continuation support at 4, 8, 12, and 16 roots. The corresponding start/positive counts were 12,971/4, 6,479/17, 4,315/21, and 3,233/22. These are event-support upper bounds, not action effects, because the immutable L2k continuation reassigns every four roots. A separate eight-episode L2d mask audit measured action-weighted persistence of 82.42% at 12 roots and 78.13% at 16. Twelve roots is therefore the shortest measured horizon near event saturation: extending to 16 adds one upper-bound positive while losing about a quarter of group starts and reducing mask persistence.

L2l freezes this one change in `experiments/l2l-stage4-action-exposure-v1.json`. Its decision unit is a uniform group-start assignment to the complete shield-bounded intention protocol. Physical HIT within 12 contiguous unit-frame transitions and before another randomized assignment is positive; an exact step-zero through step-eleven no-HIT group is negative. Observation gaps, initial non-execution, reassignment before outcome, incomplete no-HIT groups, and episode-end truncation are censored. Shield override and fail-closed input release remain recorded mediators; neither is relabeled as intended-action execution. This is an intention-to-treat factual estimand, not an invented successor.

The L2l pilot fixes two serial normal-speed natural-RNG complete Stage 4 episodes, both pilot-train-only. Each needs at least 1,800 eligible complete groups; every action needs 100 aggregate assignments; no-override and full 12-execution fractions must each be at least 70%; control dead ends remain at most 0.5%; and the exact metadata, zero-Bomb, infrastructure, linkage, and complete-episode gates remain unchanged. Sixteen accepted HIT-positive group starts is a readiness diagnostic, not a retry or contract gate. Passing with support selects only a separately frozen serial training collection. It does not admit a fit, parallel Wine, value learning, a canary, or a gameplay claim.

9.5 E4: minimal offline policy improvement

E4 begins only after E3 identifies one frozen portable representation. It compares BC with one offline expected-HIT method, initially IQL, on the same decision rows. Offline validation reports proper cost-prediction scores, calibration, action support, and episode-bootstrap intervals. Promotion then requires normal-speed original-Wine complete routes, zero Bomb, passing infrastructure contracts, lower HIT count under a preregistered serial comparison, and reported NMNB rate. Exact paired claims are allowed only if a separate exact-prefix replay gate passes; otherwise independent complete-route episode intervals are used. No offline estimate alone can promote a scorer.

9.6 E5: collection equivalence

Parallel workers are enabled only after E0. Serial and parallel fixed-seed runs must first agree on factual/action traces before independent natural-RNG collection begins. The first declared collection policy is an immutable 20% mixture of uniform exploration over the observed-shield set and the generic reactive baseline, with complete propensities and independent policy RNG streams per episode. Clock acceleration is a distinct hypothesis and cannot produce evidence until it passes an unchanged-retail-clock differential contract.

At commit `76782a4f37e0d12a9a2384561b53e68ceaf998ae`, the first declared shared-host test used two workers with eight logical CPUs each, disjoint game/controller partitions, retail frame multiplier 1.0 with coherent capture suspension, fixed retail seed `0x1234`, and the immutable 20% uniform-shield policy. The outer process tree used `SCHED_OTHER` nice 19 and idle I/O priority so normal-priority background proof jobs retained scheduling precedence. The exact command was:

```
ionice -c 3 nice -n 19 taskset -c 32-47 \
  .venv/bin/python scripts/gate_parallel_wine.py \
  --pool reference/wine-workers-v2-zklean-safe/pool.json \
  --policy-plugin src/th06_rl/policies/uniform_shield_exploration.py \
  --policy-state config/uniform_shield_exploration.json \
  --practice-stage 4 --diagnostic-rng-seed 0x1234 \
  --artifact-root artifacts/parallel-gate-exp02-76782a4-zklean-safe-nice19 \
  --corpus-root corpora/parallel-gate-exp02-76782a4-zklean-safe-nice19 \
  --output artifacts/parallel-gate-exp02-76782a4-zklean-safe-nice19/gate.json
```

The retail executable, native kernel, policy plugin, policy state, and worker pool SHA-256 digests were respectively `9f76483c46256804792399296619c1274363c31cd8f1775fafb55106fb852245`, `d26439f01400121a99137ff7d411b4bb6a2ff480daaaf1195e85a1b6cb671673`, `c70a2ad0a03bb189a01baabb62f6f404f6c4c91aa0e16c86645bfc0925f3e805`, `a63fbbb0b6d5886d327eb28b4b3d683cc13c6c759bffde8f9b2f554649068ef1`, and `76fe8c634855a2a9f11f4a5b83ab0c3498eb914df384ed40f206f199750447ad`. The pool and audit schemas were `th06-rl-normal-speed-wine-pool-v2` and `th06-rl-infra-audit-v2`.

Run	Frames	Transitions	HIT	Gaps (%)	Successors	Shield
Serial 0	25,106	25,105	16	0.0159	24,125	62
Concurrent 0	23,154	23,153	10	0.0346	22,535	64
Concurrent 1	25,455	25,454	13	0.1729	24,617	64

Table 5: E5 fixed-seed differential. Every row individually completed with zero Bomb, successor mismatch, unsafe divergence, conservative divergence, drop, callback failure, integrity error, or leftover process.

The run IDs were `20260821T101125Z-298490100`, `20260821T102110Z-520526700`, and `20260821T102110Z-450384400`; their audit SHA-256 digests were `539307713 8d1a769e12a703e35e3cae11201e79859529acfcf52b40231153366`, `8a07cabb6ca93b897853c79ede1fc7cd24151f0a356b8390981cb2d498ba8aaf`, and `8bfaff06bc25ec808f258cc0e5d1b576627741e5b53928461b639fdd43035e4d`. Capture p99 was 8.420, 8.340, and 8.692 ms; solve p99 was 0.815, 0.784, and 0.895 ms. Thus every run passed its individual latency and semantic checks, but both HIT count and normalized factual digest failed exact equality. The gate rejected `concurrent-worker-0` and emitted no admission ledger. This is a negative E5 result, not permission to weaken equality: parallel collection remains disabled.

A later read-only, record-level diagnosis separated digest noise from the first gameplay boundary. The historical normalizer omitted the run-local `snapshot_id`, and its first transition mismatch was only a coherent capture-attempt count. After ignoring only those diagnostic fields, the two concurrent traces matched the serial reference through 1,011 and 1,017 transitions. Each then advanced two retail frames where the reference advanced one. The resulting observation gap removed the one-frame execution witness and changed player position by 4 pixels; later trajectory and HIT differences are downstream. RNG, policy distribution, shield, action, and physical facts matched before that point.

The controller polls the game-frame counter before calling `NtSuspendProcess`; there is no in-process completed-root handshake. Scheduler delay in that window can admit another retail update. Normal- priority serial L2k episodes still contained nine and three rare observation gaps, so changing `nice` alone is not a proof-quality repair. Exact parallel collection remains rejected. L2l deliberately uses one serial worker rather than adding a native handshake before action-target feasibility is established.

An earlier `SCHED_IDLE` attempt is retained as a scheduling negative: its serial run completed with zero semantic mismatch but 883 observation gaps (3.62%), above the unchanged 0.5% limit. Its audit digest is `00a5a1f8fab26fac909c4f0d421ba9ece13c418b25bc81ccdefdfc9c2a68da12`.

9.7 E6: frozen-batch interaction

E6 is admitted only if E4 records a specific coverage failure or an improved candidate whose remaining error is support-limited. One immutable policy is used for each complete Wine run; parameters never update or reload within a run. The first additional collection comparator is the declared 20% uniform observed-shield mixture. Any uncertainty- or novelty-guided collector is a separate later arm with a minimum action-probability floor and the identical Wine interaction budget. All episodes remain complete, immutable, and linked to their behavior distributions. E6 was not run.

9.8 E7: transfer

E7 replaces only the environment adapter and configuration and reports unchanged versus changed code/schema surfaces, zero-shot behavior, and fine-tuning sample cost. It was not run.

This documentation checkpoint follows E0, completed E2, and negative E3 and E5 gates. E3 collected its complete serial Stage 4 inventory and produced the initial fit plus two optimization-only follow-ups. The initial model improved held-out likelihood but failed calibration and was never executed online. The bounded diagnosis is complete; offline-only L1b improved held-out metrics but did not converge within its frozen timebox and is inconclusive. L1c then converged and failed the unchanged calibration gate. The read-only residual diagnosis selected one small current-observation MLP without reading scaled validation. That model converged and improved held-out proper scores but failed calibration. The completed target-contract diagnosis found exact propensities and localized the residual to flat-model misspecification after its other frozen explanations missed their materiality thresholds. The selected shared-action scorer then failed to converge and was worse than L1d on both splits, exposing additive-score mismatch with the lexicographic rule. The current-observation execution/dynamics and factual-risk pilot then passed its frozen diagnostic gate at supported horizons. This establishes portable current-root factual signal but neither causal action value nor an authorized learned candidate. The state-only reused-heldout attribution found weak incremental action-relative HIT signal at 16 frames, none at 64, and robust incremental shield signal.

The subsequent support/lifecycle decomposition localized that weak HIT increment to baseline-equal actions: it reversed sign on actions known to differ from the reactive baseline, while pre-first-HIT rows still favored full. This rejects a pure post-HIT explanation but not reactive route/tie-breaking context. Eight fresh episodes independently confirmed the overall 16-frame proper-score increment and pre-first-HIT direction. The nonbaseline point gate also passed, but only three of eight episode directions favored full, and calibration readiness failed; value learning remains unadmitted. The subsequent train-only Platt fit then passed the global calibration limits but worsened Brier against both frozen comparators and in the nonbaseline and pre-first-HIT views. This rejects scalar calibration of the frozen ridge score without weakening the confirmed predictive-signal claim. The aligned direct Brier model then substantially improved the frozen full score and the same-architecture state-only comparator while passing global calibration. Its formal gate still rejected fresh confirmation because the nonbaseline interval narrowly crossed zero and pre-first-HIT episode directions were heterogeneous. A propensity-stratified diagnosis localized the missing increment to the bulk of randomized actions below probability 0.025, whereas baseline-heavy rows improved in every episode. The next frozen change is therefore bounded uniform-shield-target weighting. That exact weighting test then worsened target Brier against frozen L2f and failed every action-signal view, pre-first-HIT stability, and its candidate-bound gate despite passing weight and calibration contracts. Logged action measure is rejected as the primary root cause. Fixed short portable history is now the selected representation test; value learning remains unadmitted.

10 Threats to Validity

The shield horizon may be too short, and unknown births may violate its narrow claim. Expected HIT count is only a surrogate for NMNB probability and may rank nonzero-HIT policies differently. HIT continuation changes later physical occupancy through lifecycle, resources, rank, bullet clearing, and invulnerability; complete-route training and survival-conditioned diagnostics must therefore be reported separately. Behavior action support does not guarantee support for histories reached by a stronger no-HIT policy, and a positive propensity floor does not prevent long-horizon OPE variance.

Complete-route episodes may be too few for stable episode-level intervals, future-HIT prediction, or uncertainty estimates. Wine scheduling may introduce observation gaps; command pickup may be mistaken for a new policy decision; and Touhou 6–specific memory facts may leak through an apparently generic feature builder. E0 and E2 are therefore prerequisites, not housekeeping. The transfer claim is falsified if downstream learner, corpus, objective, or evaluator semantics require title-specific branches.

11 Artifact and Reproducibility Contract

The implementation uses repository-relative scripts. The LaTeX artifact is built by `scripts/check_paper.sh`. Runtime setup, native compilation, Wine execution, corpus validation, and route verification have distinct entry points. The canonical compiled paper is tracked at `paper/main.pdf`; intermediate LaTeX files stay in the ignored `paper/build/` directory. Wine prefixes, corpora, and fitted models remain untracked outputs. Committed experiment claims must point to immutable hashes rather than mutable filenames.

12 Conclusion

The project asks how much complete-route HIT reduction can be learned from portable physical history when exact Wine interaction is costly, input pickup is delayed, future births are partially observed, and the online filter covers only instantiated hazards. The goal is NMNB; the first optimization claim is strictly expected undiscounted HIT reduction. The initial learner has three parts: an audited decision-epoch view, transparent BC, and immutable shield-constrained Wine evaluation. Every additional representation, algorithm, objective, uncertainty mechanism, or collection strategy must solve one measured failure of that baseline.

E0 establishes the serial factual data plane, E2 establishes the causal learner projection and first exporter, and E5 rejects the current parallel pool. E3 completed its serial Stage 4 inventory and initial frozen supervised fit; held-out likelihood improved but calibration failed, so it emitted no online candidate. The diagnosis localized the next test to convergence of that same transparent fit. L1b improved held-out likelihood and calibration without a representation change but exhausted its optimizer timebox before the frozen gradient gate. L1c changed only that ceiling, reached the train-only gate, and still failed held-out calibration. The linear current-observation candidate is therefore rejected under the joint gate. Train-only scale correction was insufficient, current features exactly reconstructed the collector, and the fixed small MLP then improved held-out NLL but also failed calibration. The frozen target-contract diagnosis verified every recorded propensity, found the sampled-label, stopping, and discarded-soft-target explanations insufficient, and localized the residual to the flat model's handling of a reconstructible action-conditioned lexicographic rule. The selected shared additive scorer then exhausted its timebox and performed worse, with zero agreement on the final tie strata. The known collector distribution remains an exact data-plane control. The frozen offline current-observation execution/dynamics and factual HIT-risk pilot then passed at its supported horizons. Its next boundary is incremental action signal and calibrated/support-aware factual risk, not history or value learning. The incremental-action attribution found weak 16-frame but not 64-frame HIT evidence and worse full-model calibration; the fixed lifecycle/support attribution then found that the increment was baseline-equal and reversed sign on known nonbaseline actions. Fresh confirmation subsequently reproduced the overall signal on all eight episode directions and a weak negative nonbaseline point delta, while failing the unchanged calibration-readiness gate. The next boundary is one train-only calibration/probability model, not history or value learning. The first such test, a two-parameter monotone Platt surface, repaired global calibration but failed its proper-score and signal-preservation gates and is rejected. The aligned direct probability test then established a much stronger overall hazard model and incremental action-relative signal, but failed its frozen nonbaseline and pre-first-HIT stability gates. Its formal decision is reject, so no fresh confirmation or policy ran; its research decision was modify because the failure was concentrated in the low-propensity action measure rather than the full candidate's overall discrimination or calibration. The subsequent exact bounded uniform-shield-target weighting test rejected that proposed main cause: it worsened its target Brier comparator, did not stabilize exploratory action signal, and produced a heavily clipping raw surface. Its formal decision is also reject. The subsequent fixed unit-frame 16-root scalar history improved its point Brier and aggregate action ablation, but temporal gain was heterogeneous, exploratory and pre-first-HIT action intervals crossed zero, and the raw surface clipped materially. It too is rejected. The subsequent cap-48 primitive set improved aggregate action ablation and ranking but failed object-gain, exploratory-action, lifecycle, calibration, and saturation gates; it is also rejected. Its loss-only BCE-with-logits follow-up improved Brier, NLL, ranking, and confident missed positives, but still failed the joint gate. The decisive read-only audit found that 99.45% of h16 positives occurred only after the initial action ended, so the target is not supported current-action value. Further learner sweeps on this target are stopped. L2k then established the safe fixed-short-action-exposure collection boundary with uniform four-root intentions and explicit assignment and override provenance over two serial complete Stage 4 pilot-train episodes. Its original rejection is retained, while a separately hashed, bounded audit erratum aligned the checker with the preregistered control-dead-end lifecycle case and passed all seven gates without changing data or thresholds. First-divergence analysis kept parallel disabled at the external poll-to-suspend scheduling race, while target sizing found zero strict h4 positives and only four accepted h4 ITT positives. L2l therefore freezes a serial 12-root intention/h12 HIT pilot, the shortest measured horizon near event-support saturation. No learner has yet run and value learning remains later. No learned policy has run online, exact-prefix branching is not established, and this checkpoint makes no policy-improvement, NMNB, or transfer claim.

References

[1] Lili Chen, Kevin Lu, Aravind Rajeswaran, Kimin Lee, Aditya Grover, Michael Laskin, Pieter Abbeel, Aravind Srinivas, and Igor Mordatch. Decision transformer: Reinforcement learning via sequence modeling. In *Advances in Neural Information Processing Systems*, 2021. URL `https://arxiv.org/abs/2106.01345`.

[2] Ilya Kostrikov, Ashvin Nair, and Sergey Levine. Offline reinforcement learning with implicit q-learning. In *International Conference on Learning Representations*, 2022. URL `https://arxiv.org/abs/2110.06169`.

[3] Aviral Kumar, Aurick Zhou, George Tucker, and Sergey Levine. Conservative q-learning for offline reinforcement learning. In *Advances in Neural Information Processing Systems*, 2020. URL `https://arxiv.org/abs/2006.04779`.

[4] Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabás Póczos, Ruslan Salakhutdinov, and Alexander J. Smola. Deep sets. In *Advances in Neural Information Processing Systems*, 2017. URL `https://arxiv.org/abs/1703.06114`.